



# Fundamentals of neurobiology

## *ch2 : the action potential*

R.C. Lambert

*Neuronal networks and physiopathological rhythms*

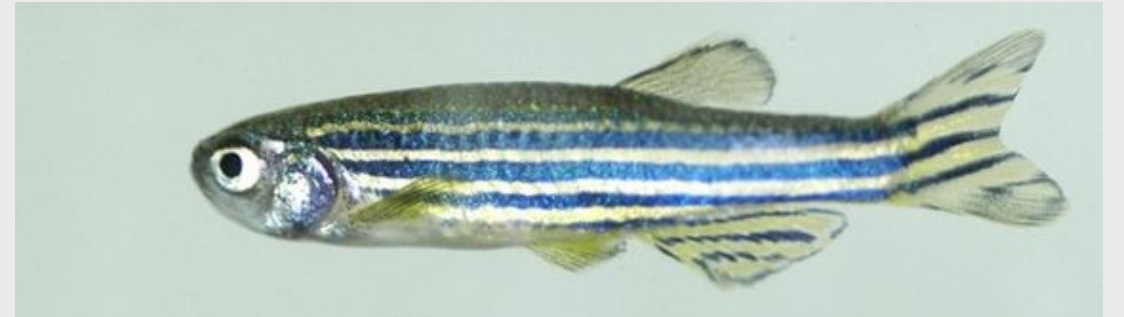
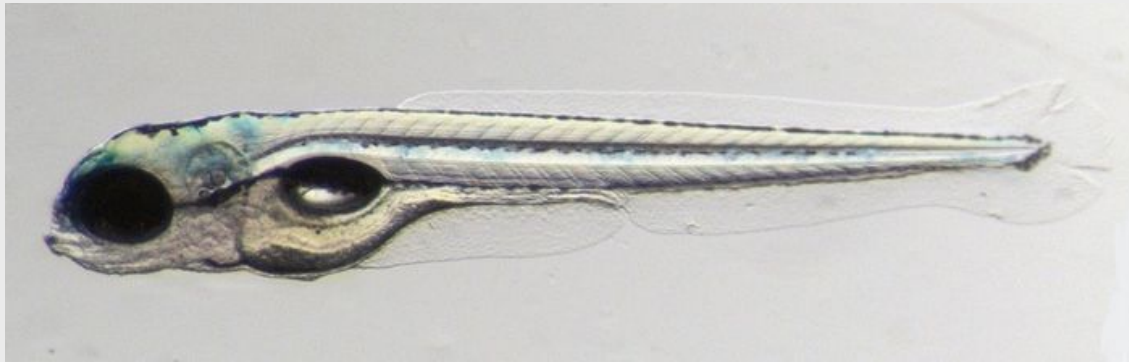
<http://www.ibps.upmc.fr/fr/Recherche/umr-8246/reseaux-de-neurones-et-rythmes-physiopathologiques>

*Sorbonne Université- Neuroscience Paris Seine (NPS)*

*CNRS UMR8246, INSERM U1130, UPMC UMI19*

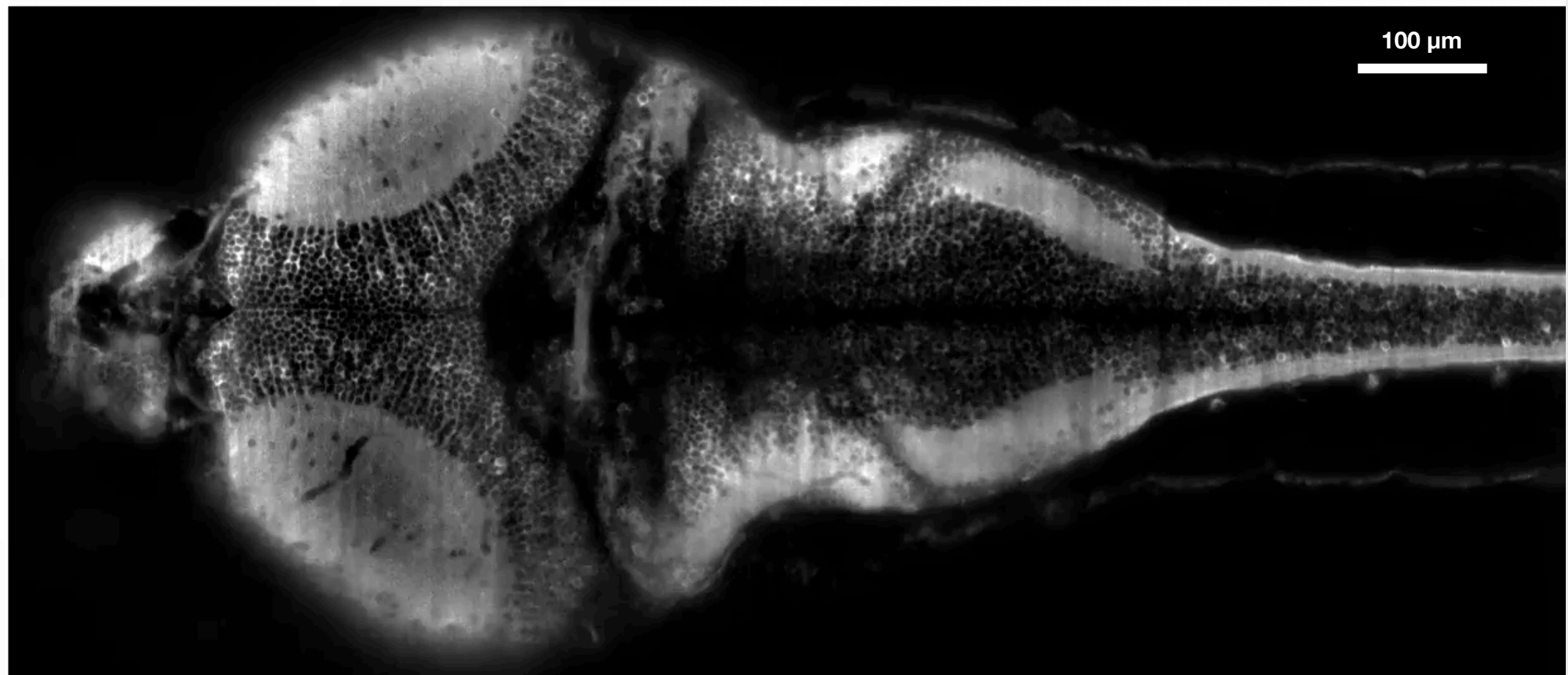


## brain computation ... electrical activity



zebrafish larvae (GCaMP3)

t = 0.0 sec



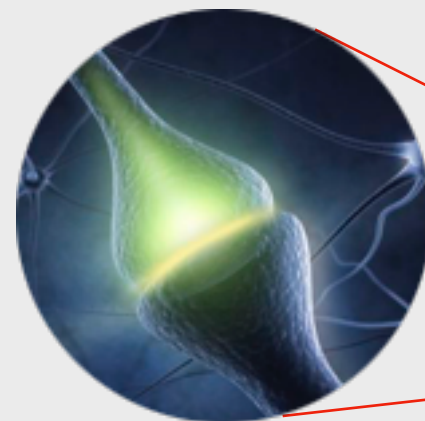
*Panier et al., 2013*

# brain computation ... electrical activity

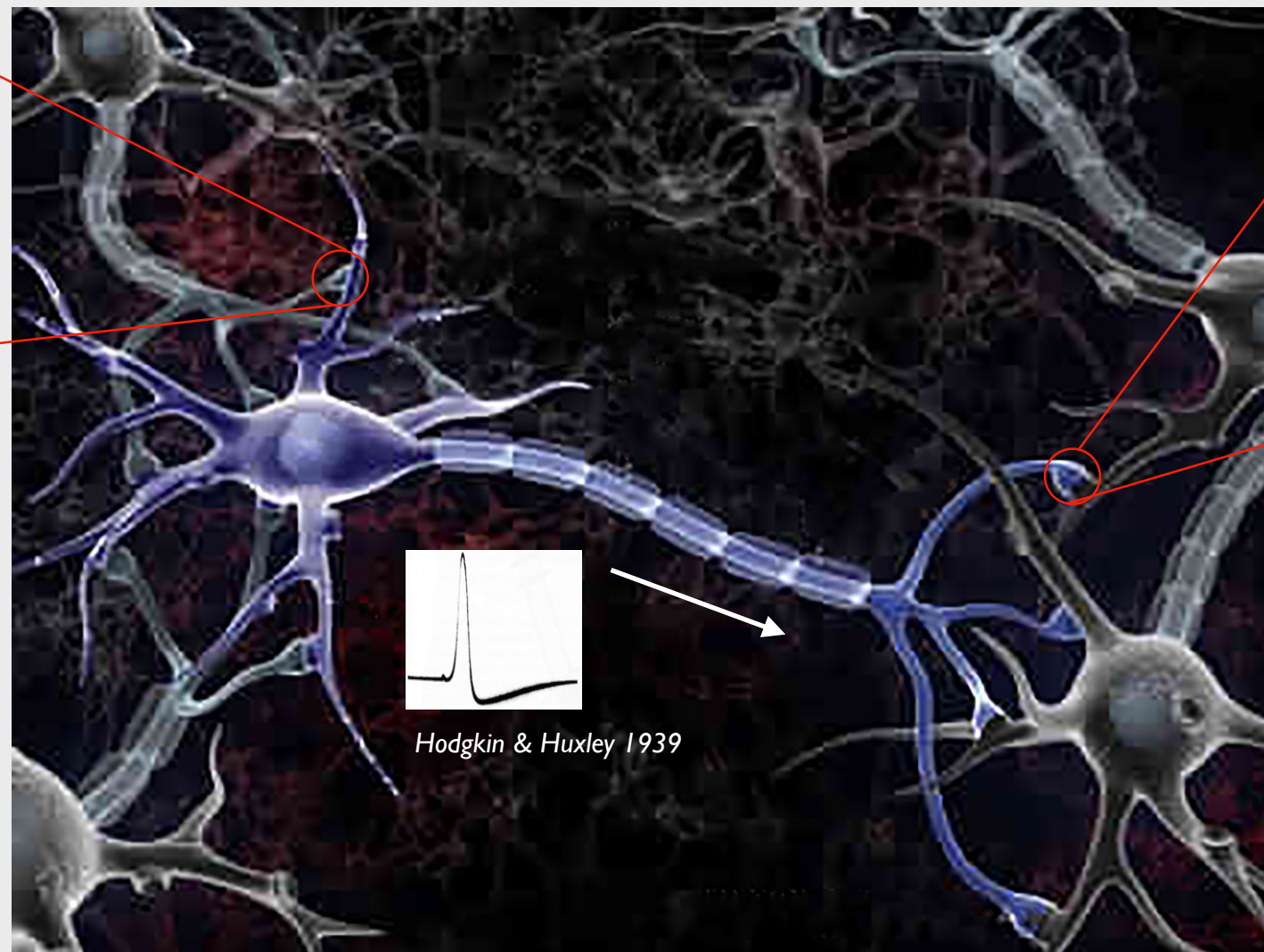
## Number of neurons in the brain

- c elegans: 302
- Fruit fly: 100 thousand ( $10^5$ )
- Cockroach: 1 million ( $10^6$ )
- Mouse: 75 million ( $10^8$ )
- Cat: 1 billion ( $10^9$ )
- Chimpanzee: 7 billion ( $10^9$ )
- Elephant: 23 billion ( $10^{10}$ )
- **Human: 86 billion ( $10^{11}$ )**

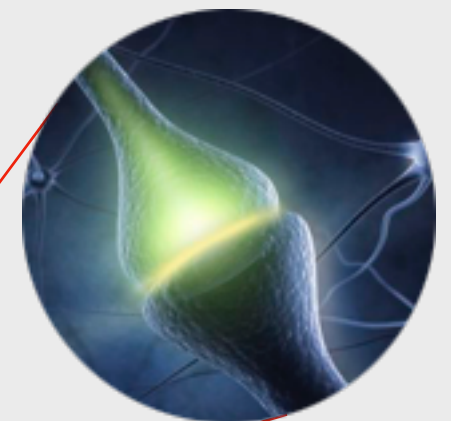
each neuron can receive several thousand connections  
(average number of synapses of a neocortical neurons : 7000)



*electrical to chemical  
signal*

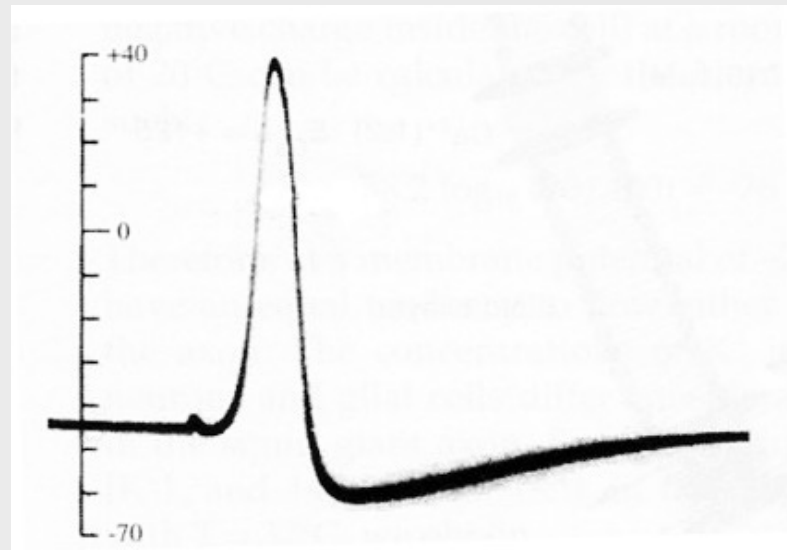


*electrical to chemical  
signal*

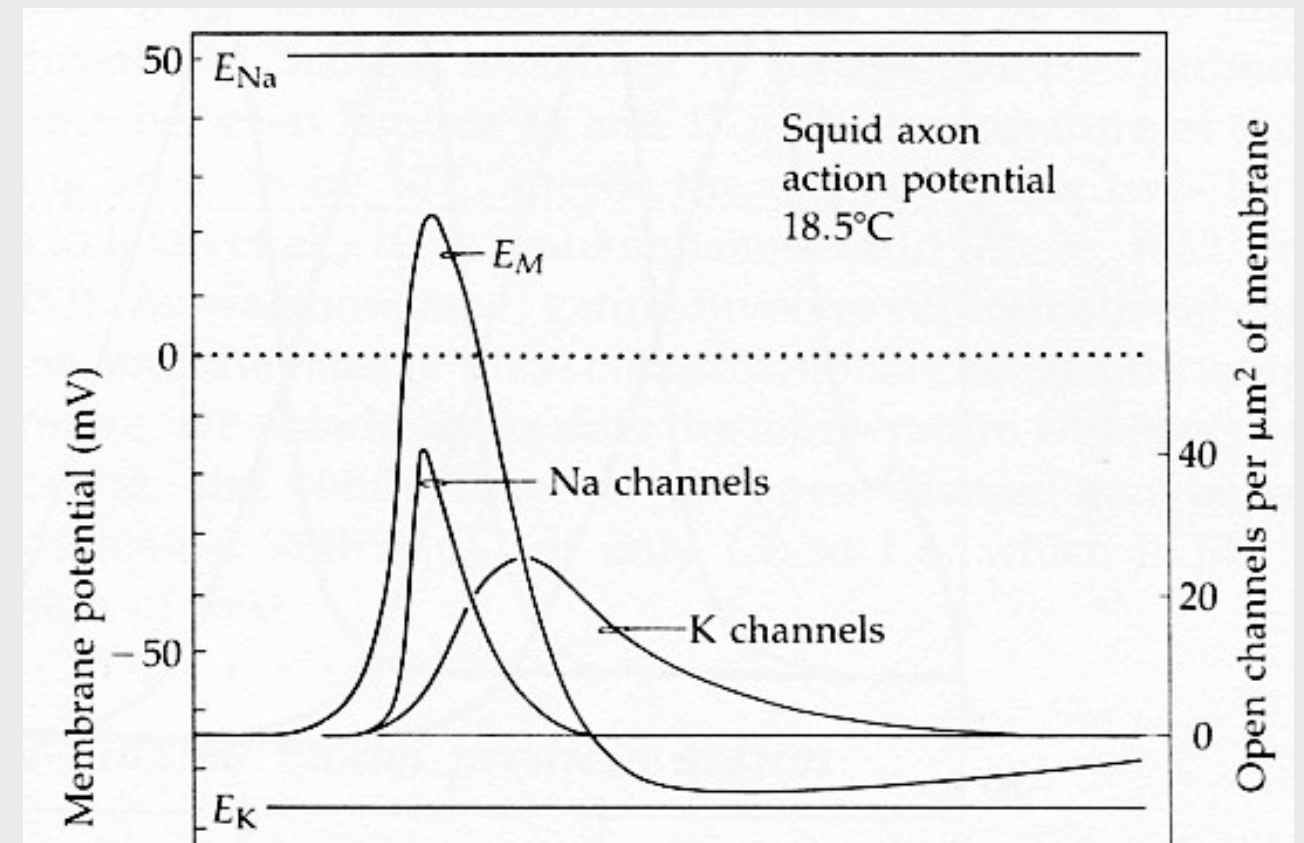


# action potential generation

squid axon



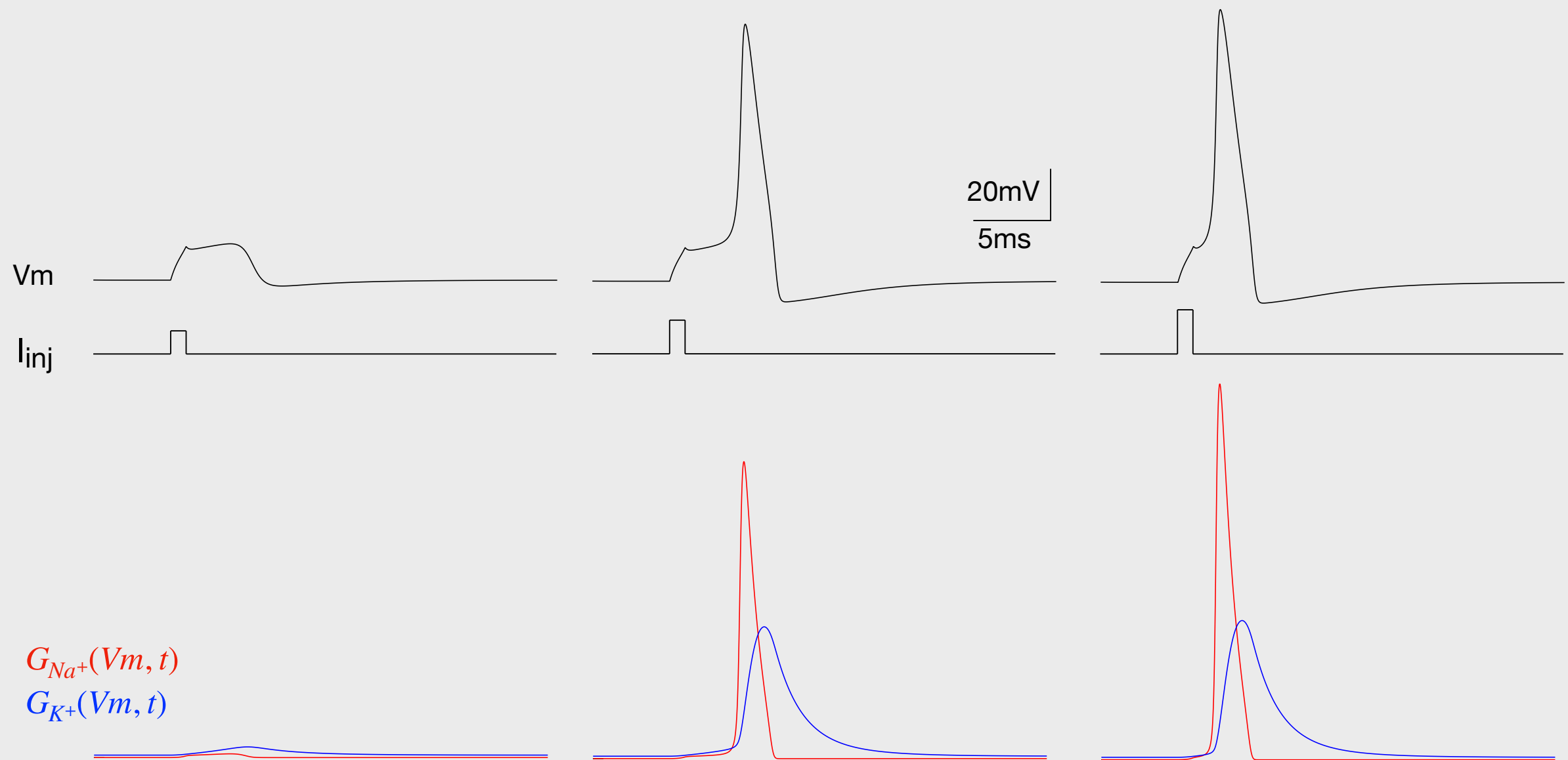
*Hodgkin & Huxley 1939*



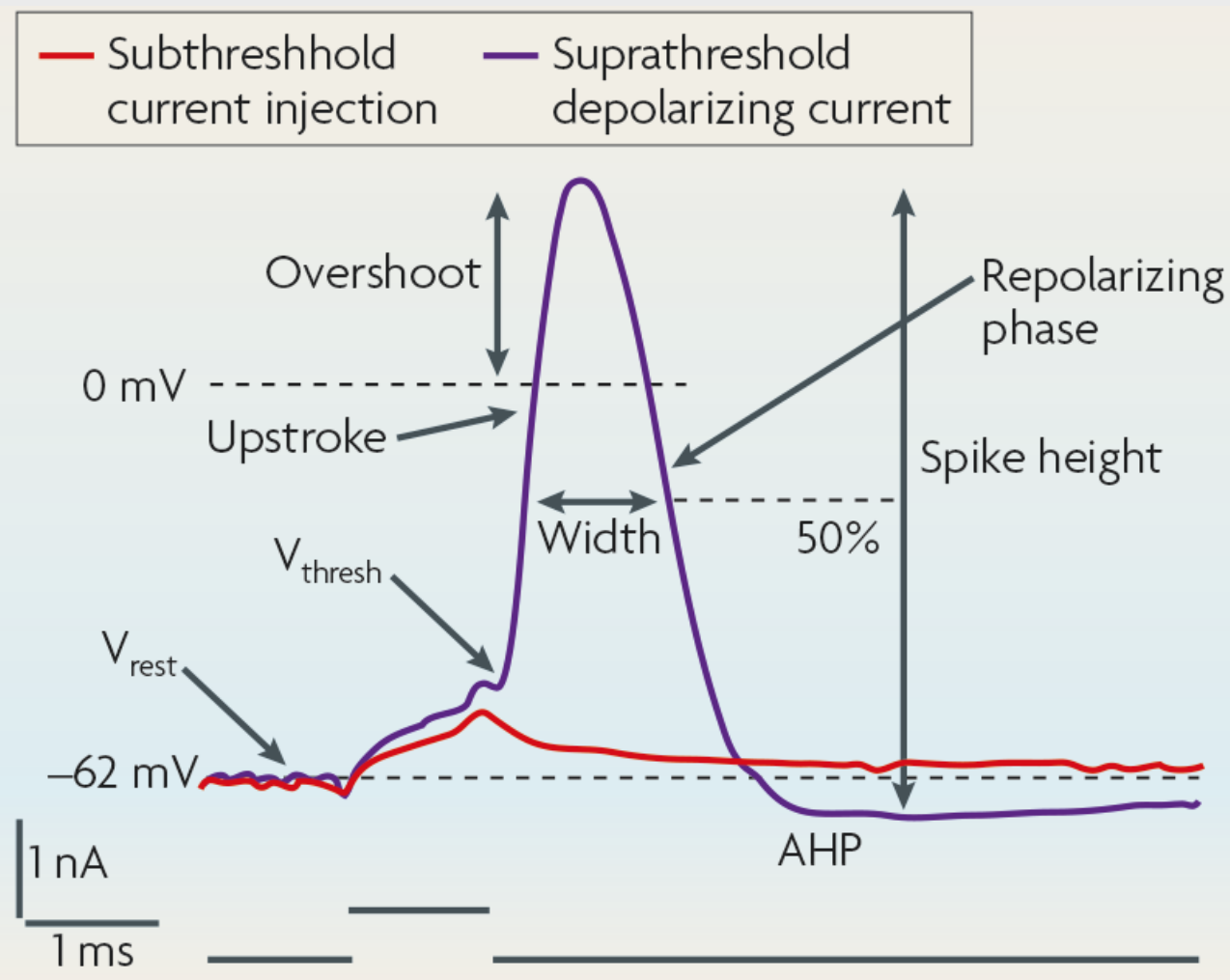
*Hodgkin & Huxley 1952*

regenerative depolarization due to a steeply voltage-dependent Na conductance followed by repolarization due to inactivation of this conductance along with activation of a slower K conductance

# action potential generation

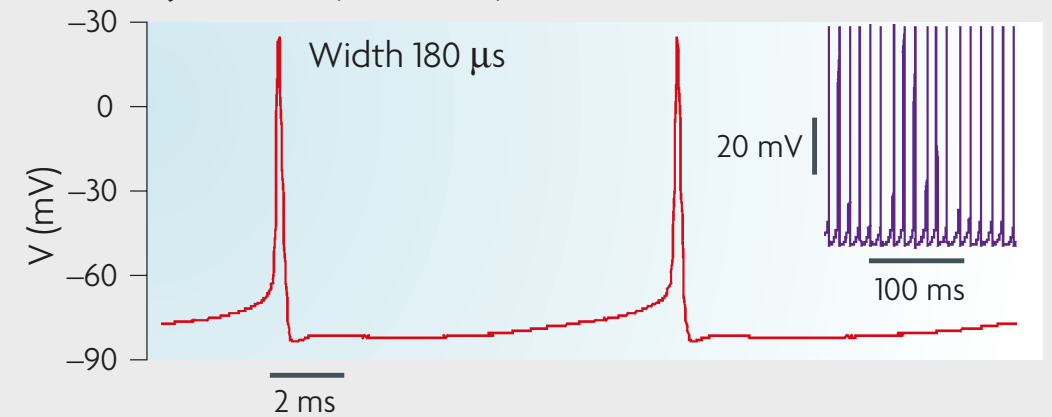


# action potential diversity

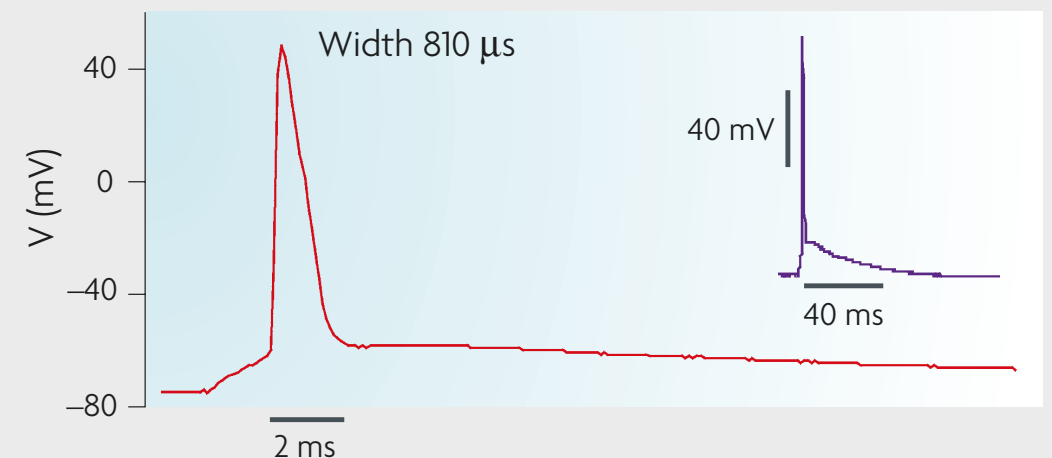


AHP: After Hyperpolarization

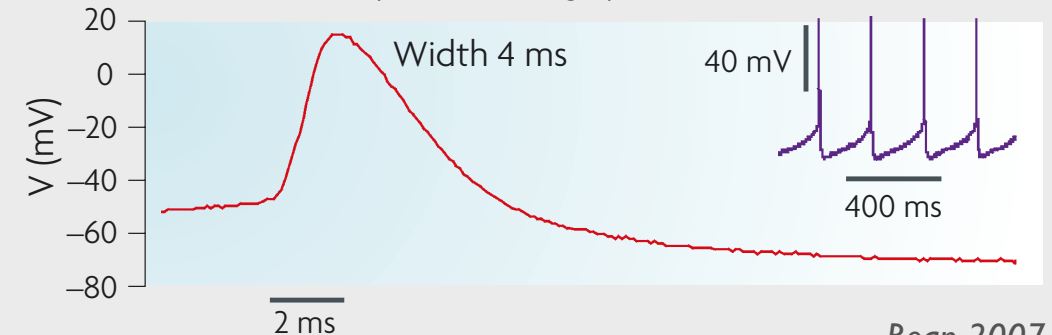
**a Purkinje neuron (cerebellum)**



**b CA1 pyramidal neuron (hippocampus)**



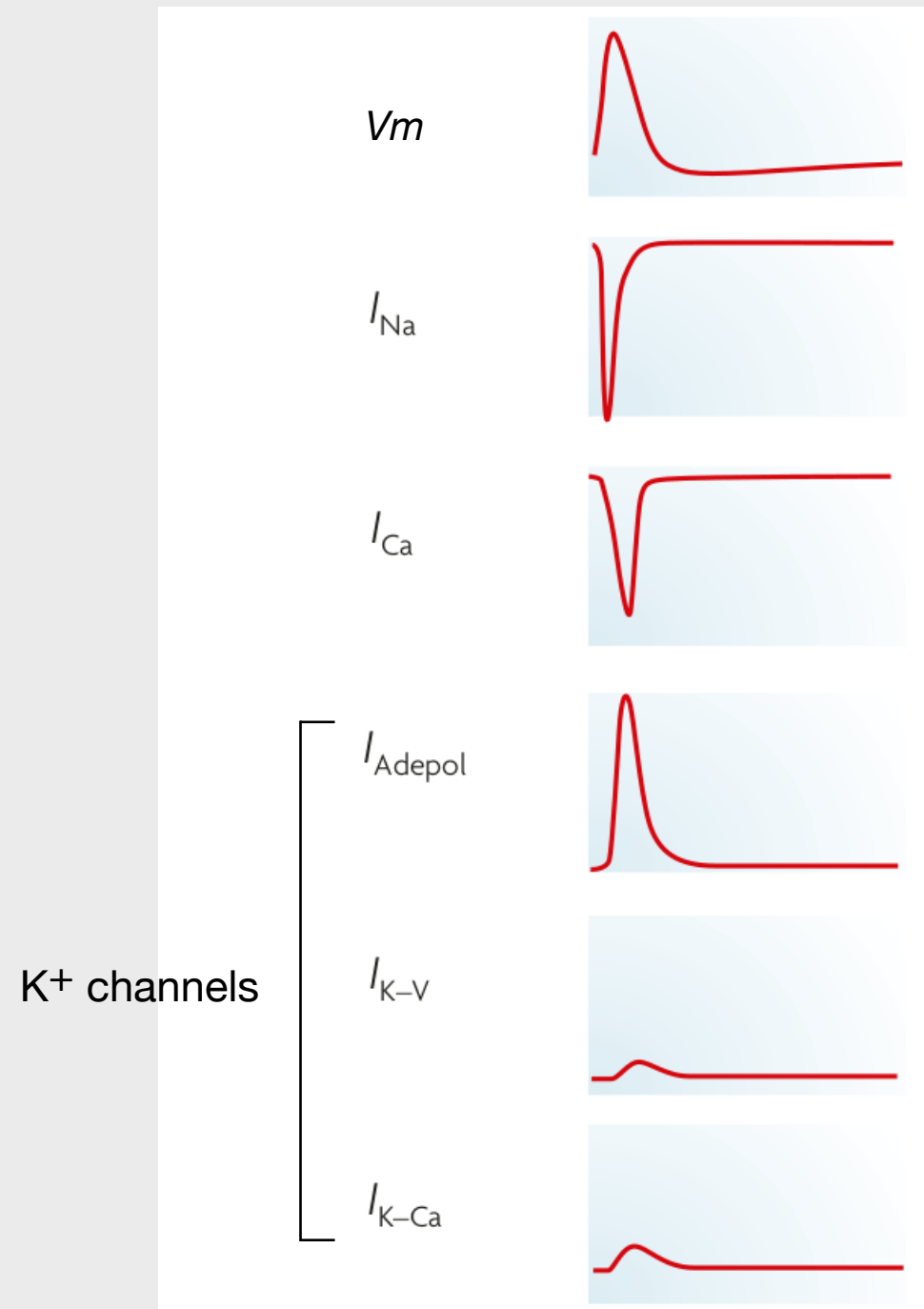
**c Dopamine neuron (substantia nigra)**



Bean 2007

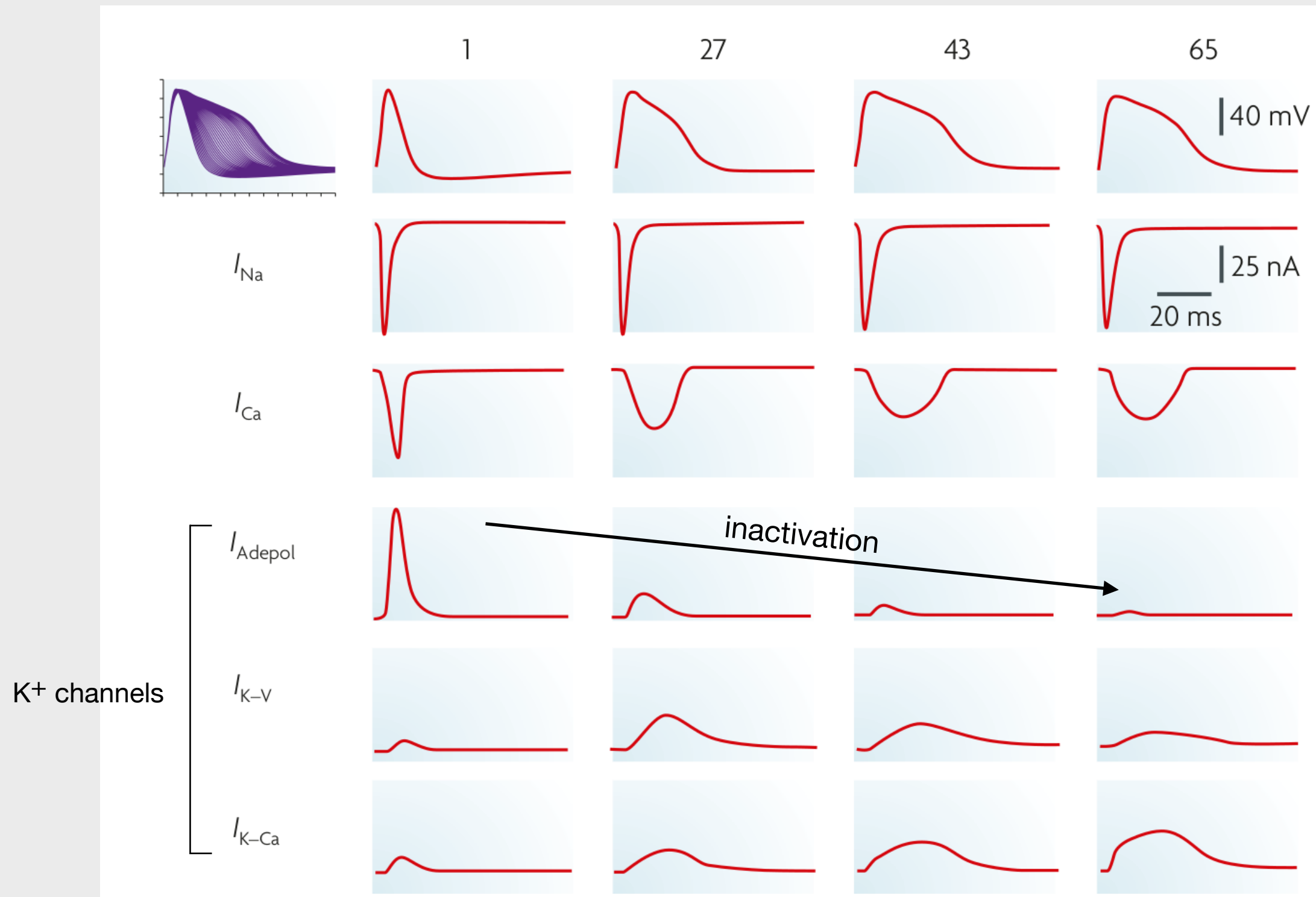
In the years since the Hodgkin–Huxley analysis of the squid axon action potential, it has become clear that most neurons contain far more than the two Na and K voltage-dependent conductances found in the squid axon

# action potential diversity



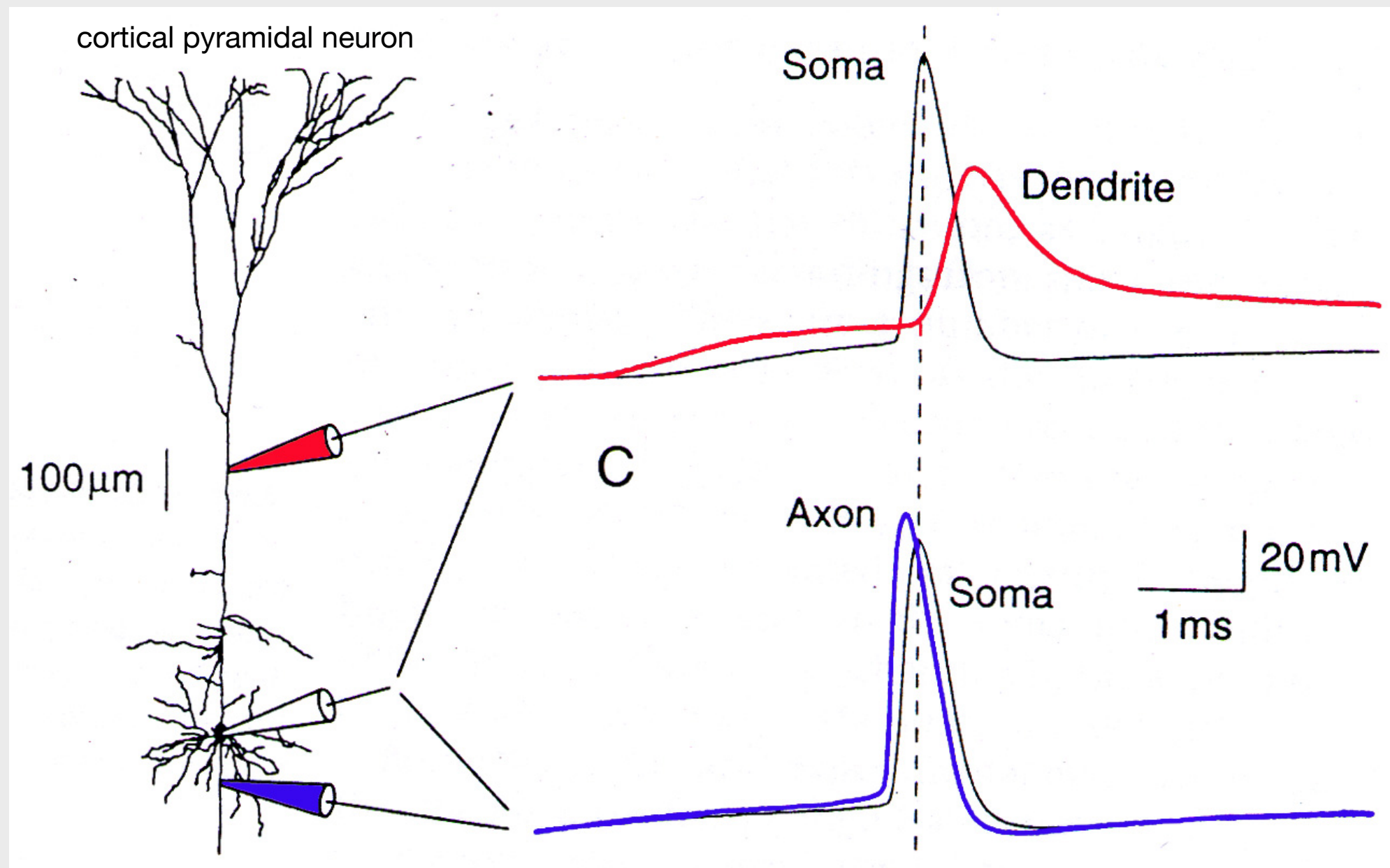
In the years since the Hodgkin–Huxley analysis of the squid axon action potential, it has become clear that most neurons contain far more than the two Na and K voltage-dependent conductances found in the squid axon

# action potential variability



the action potential becomes generally broader in response to increased frequency of firing because of the cumulative inactivation of  $K^+$  channels (here  $I_A$  which is only partially offset by the increased recruitment of  $I_{K-V}$  and  $I_{K-Ca}$ )

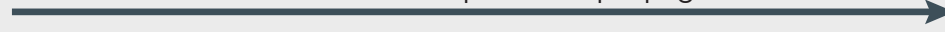
## where is the action potential created ?



# neurons are highly polarized cells

**a** Functional polarity

Direction of action potential propagation



**b** Anatomical polarity

Somatodendritic input

Axonal output

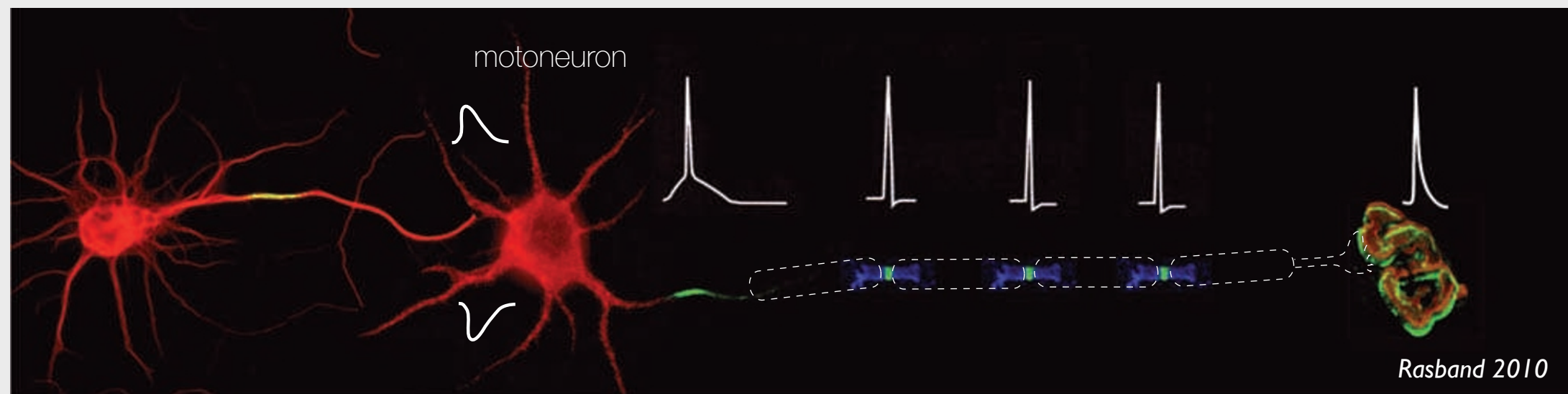
**c** Subcellular polarity

Synapse

Axon initial segment

Nodes of Ranvier

Neuromuscular junction



# AIS contains a high-density of Na channels

**a** Functional polarity

Direction of action potential propagation →

**b** Anatomical polarity

Somatodendritic input

Axonal output

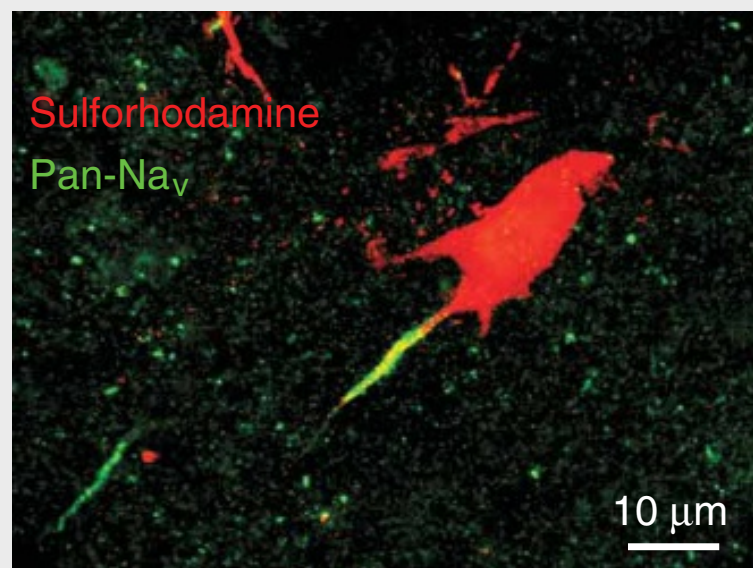
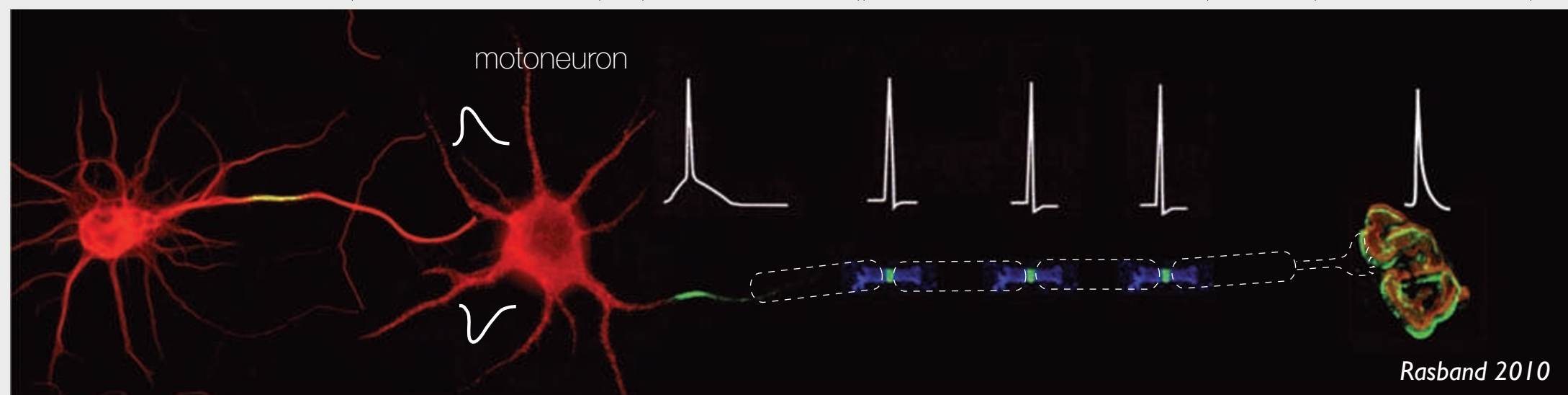
**c** Subcellular polarity

Synapse

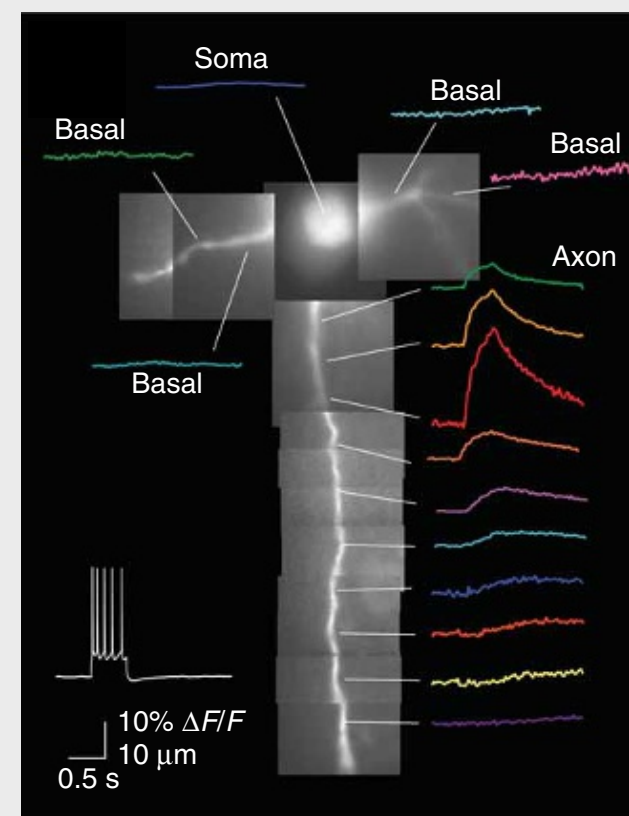
Axon initial segment

Nodes of Ranvier

Neuromuscular junction



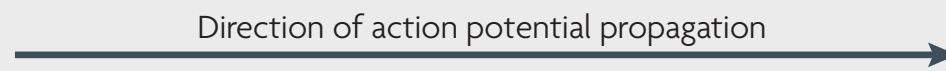
cortical pyramidal neuron filled with red dye  
+ Na channels antibody (green)



Fluorescent image of a cortical  
pyramidal neuron filled with the  
Na-sensitive dye SBFI

# AIS contains a high-density of Na channels

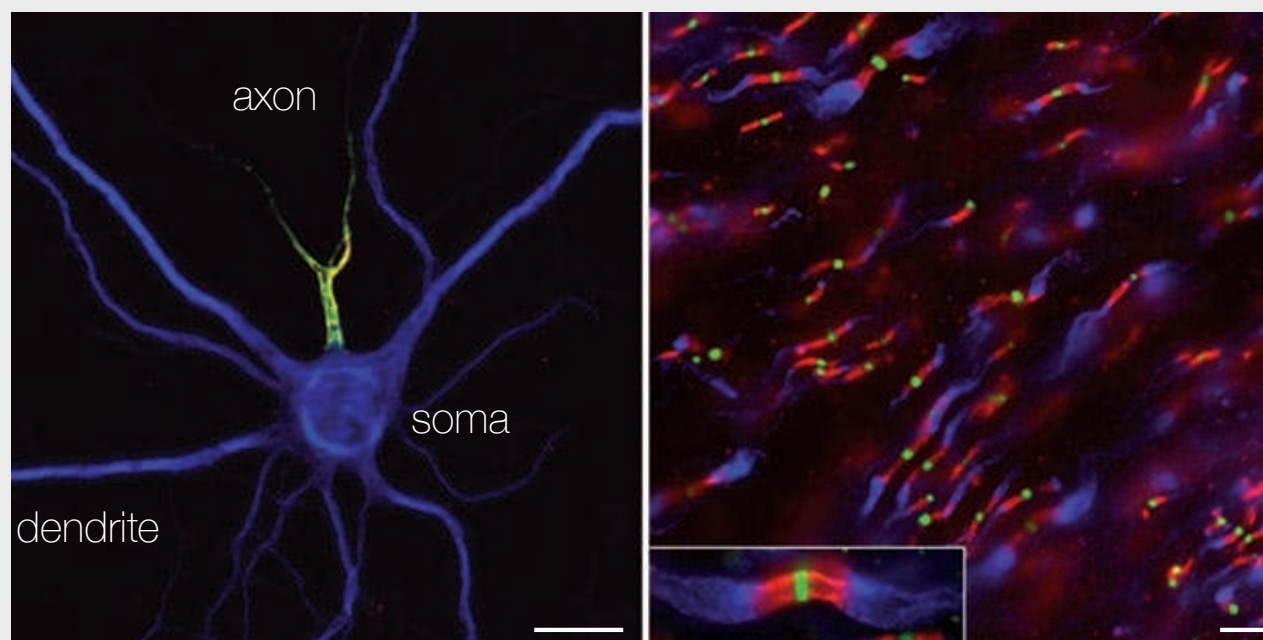
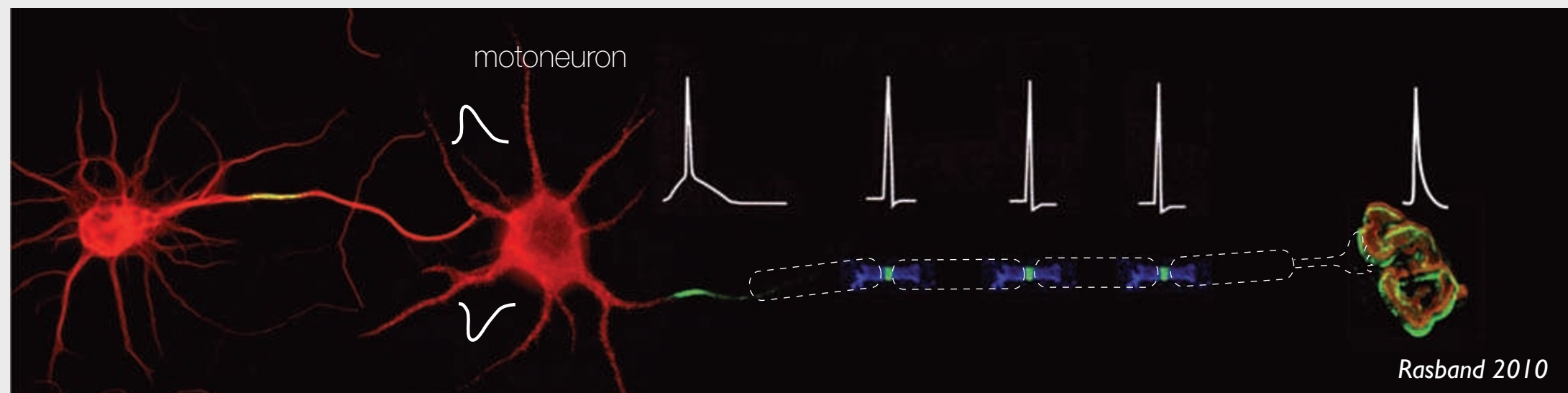
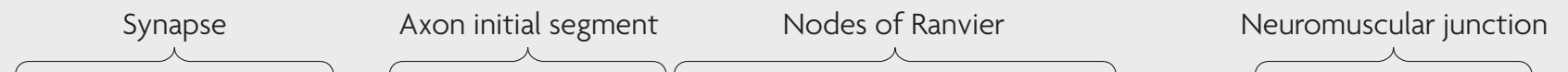
**a** Functional polarity



**b** Anatomical polarity



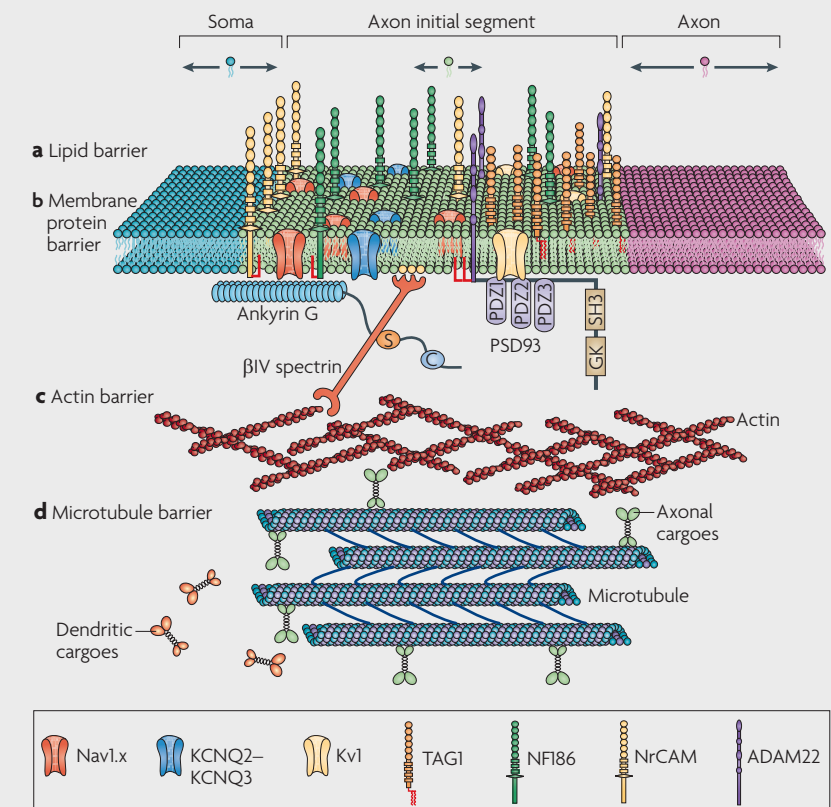
**c** Subcellular polarity



cultured hippocampal neurons

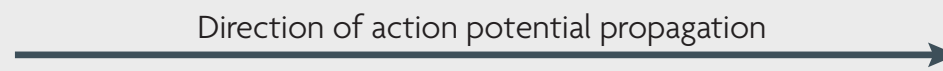
+ ankyrin G antibody (green)

nodes of Ranvier



# AIS contains a high-density of Na channels ... and Ca, K, ... channels

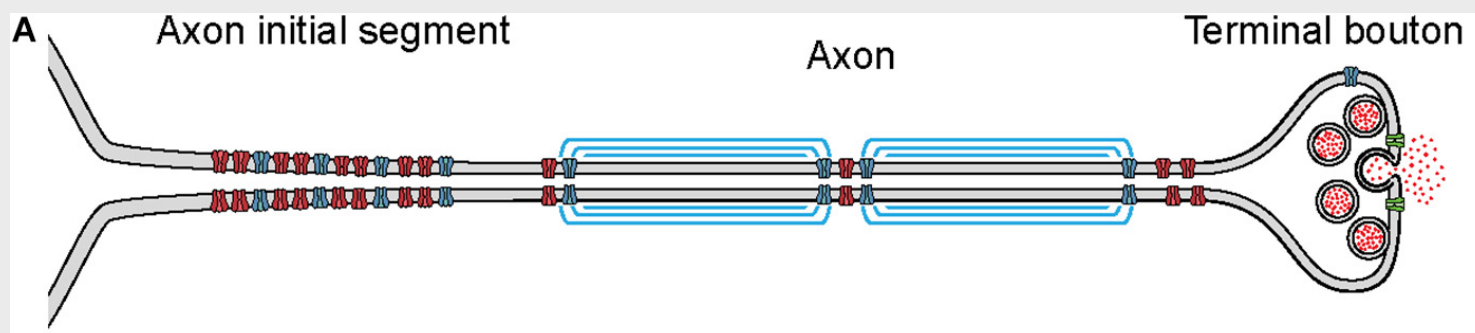
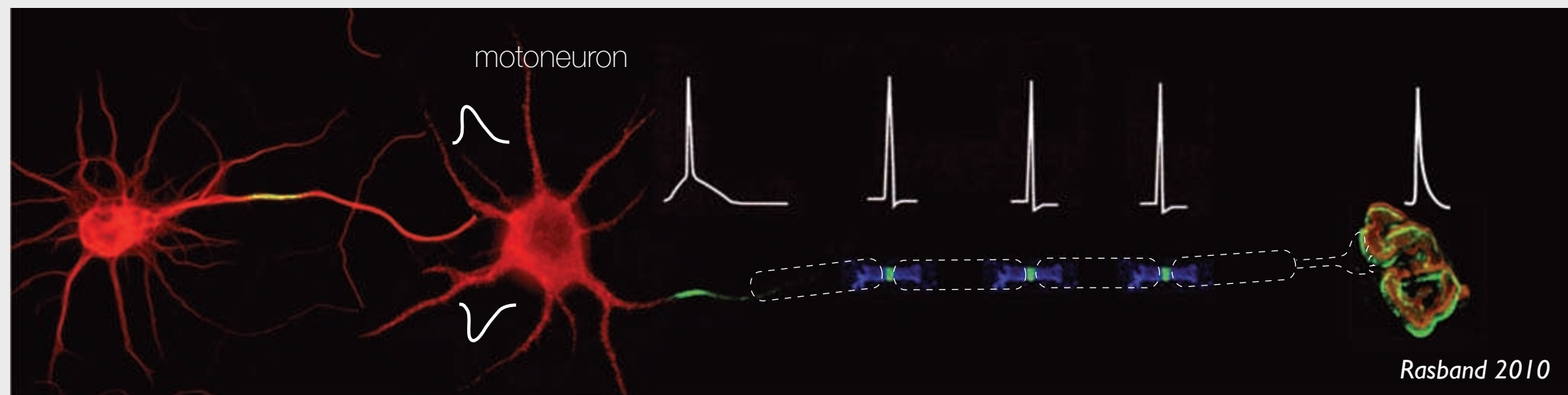
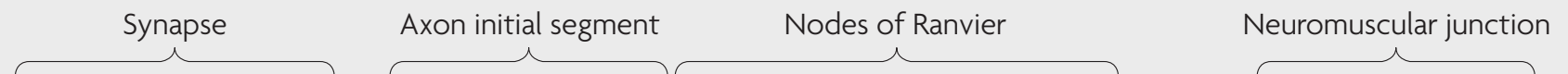
**a** Functional polarity



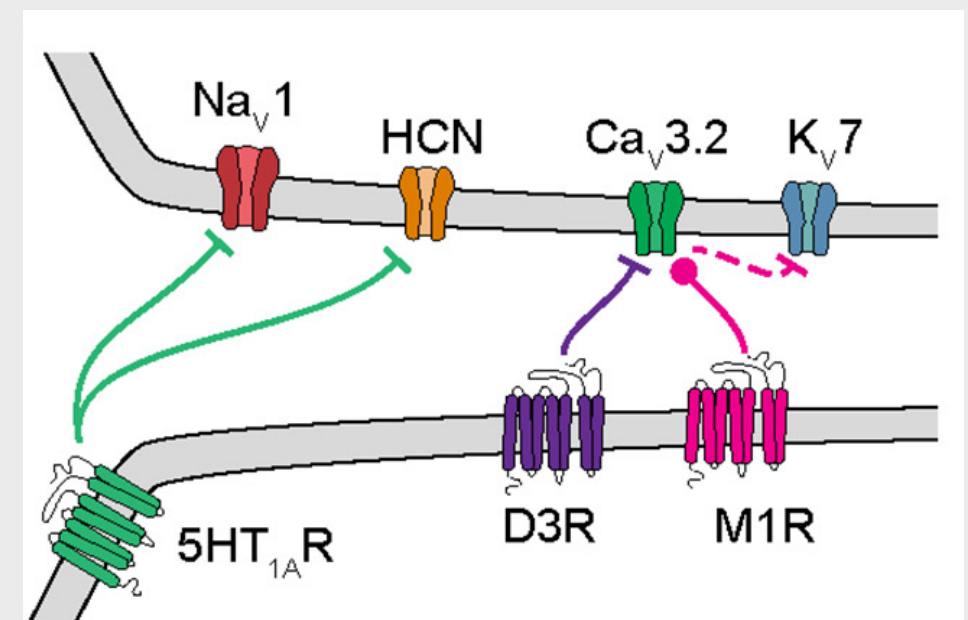
**b** Anatomical polarity



**c** Subcellular polarity



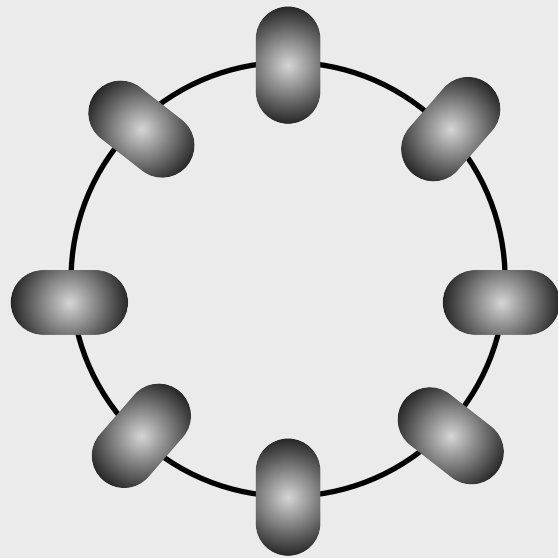
Burke et al 2019



**BREAK**

# neurons are not isopotential

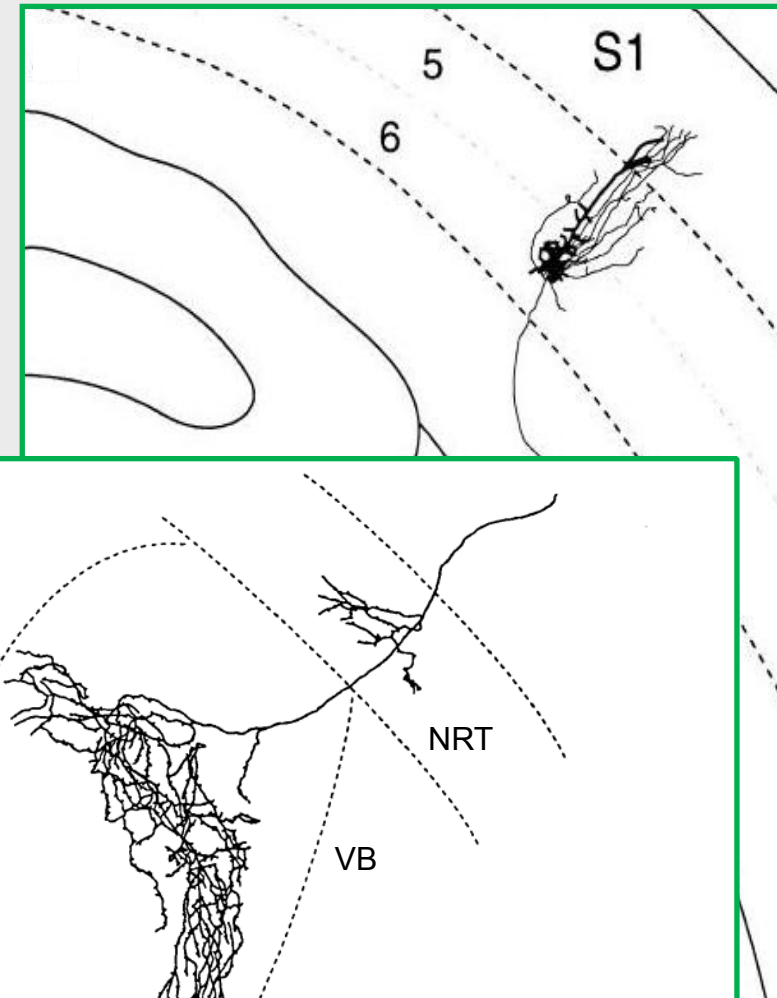
cortical pyramidal neuron



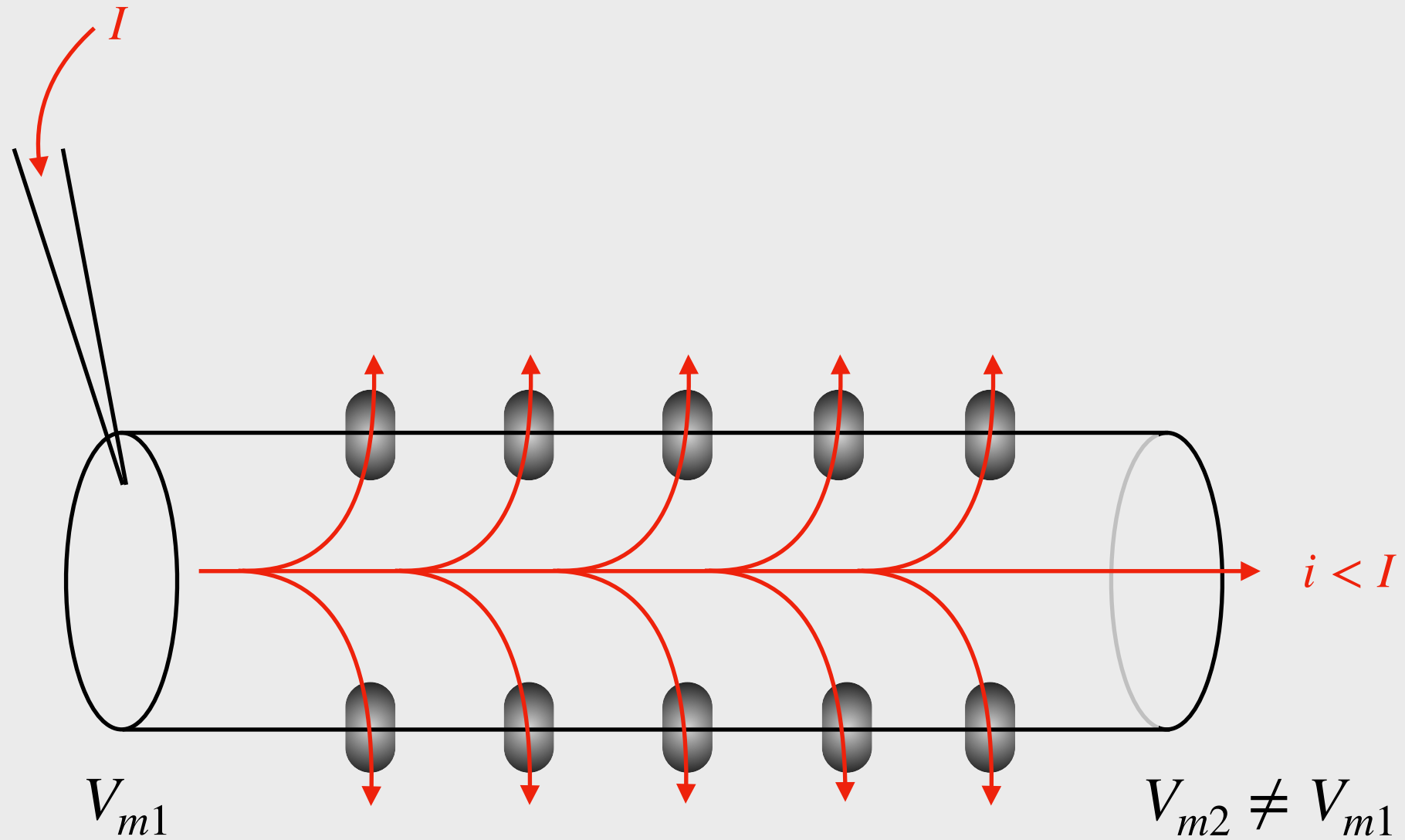
$$R_m = \frac{1}{G_L}$$

$$C_m = A \cdot c_m$$

$A$  : surface area  
 $c_m$  : specific capacitance  
 $G_L$  : leak conductance

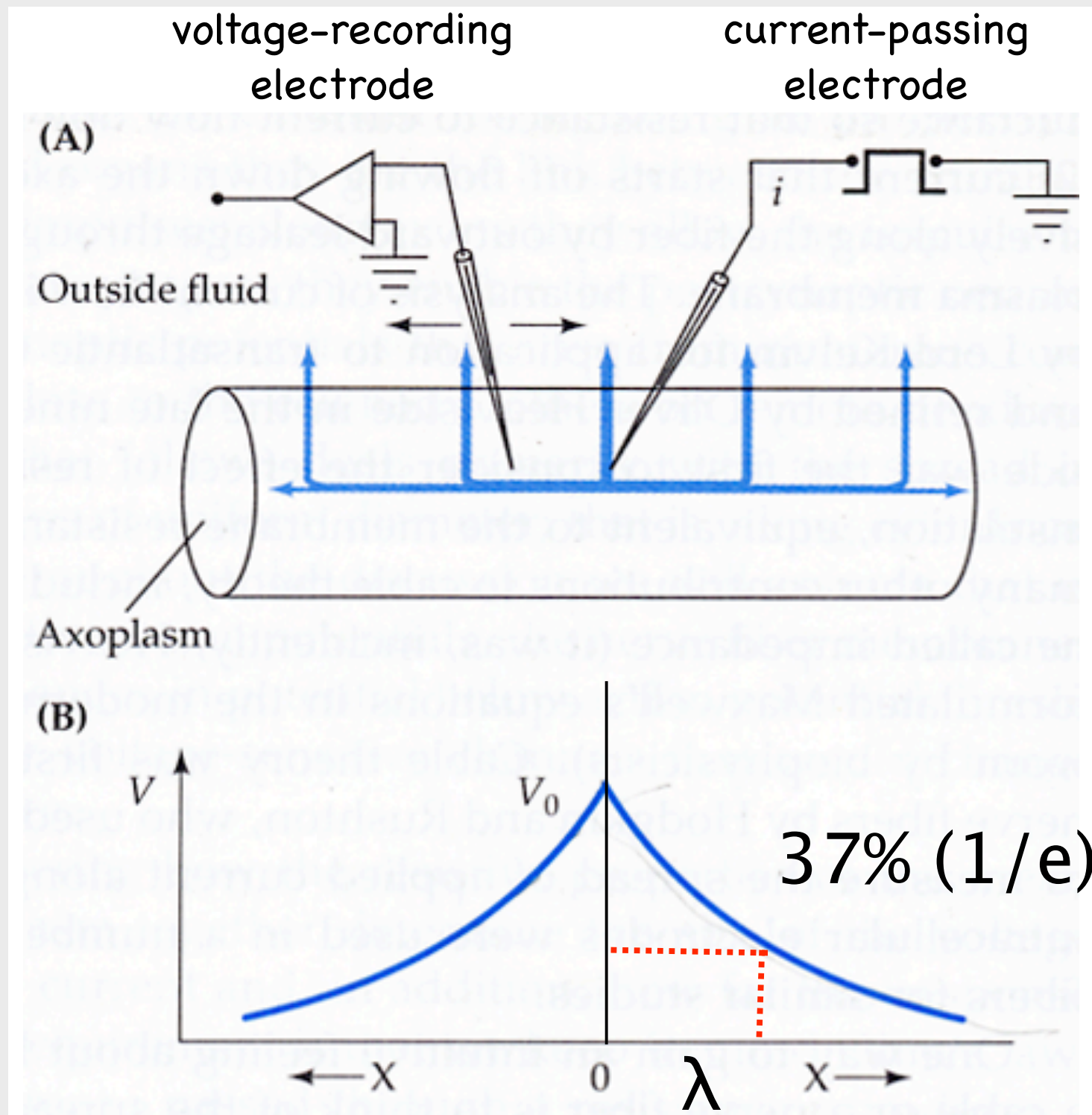


## neurons are not isopotential



the drop in membrane potential depends on the leak conductance (membrane resistance) and the ability of ions to « move » inside the cylinder (axial resistance)

## neurons are not isopotential



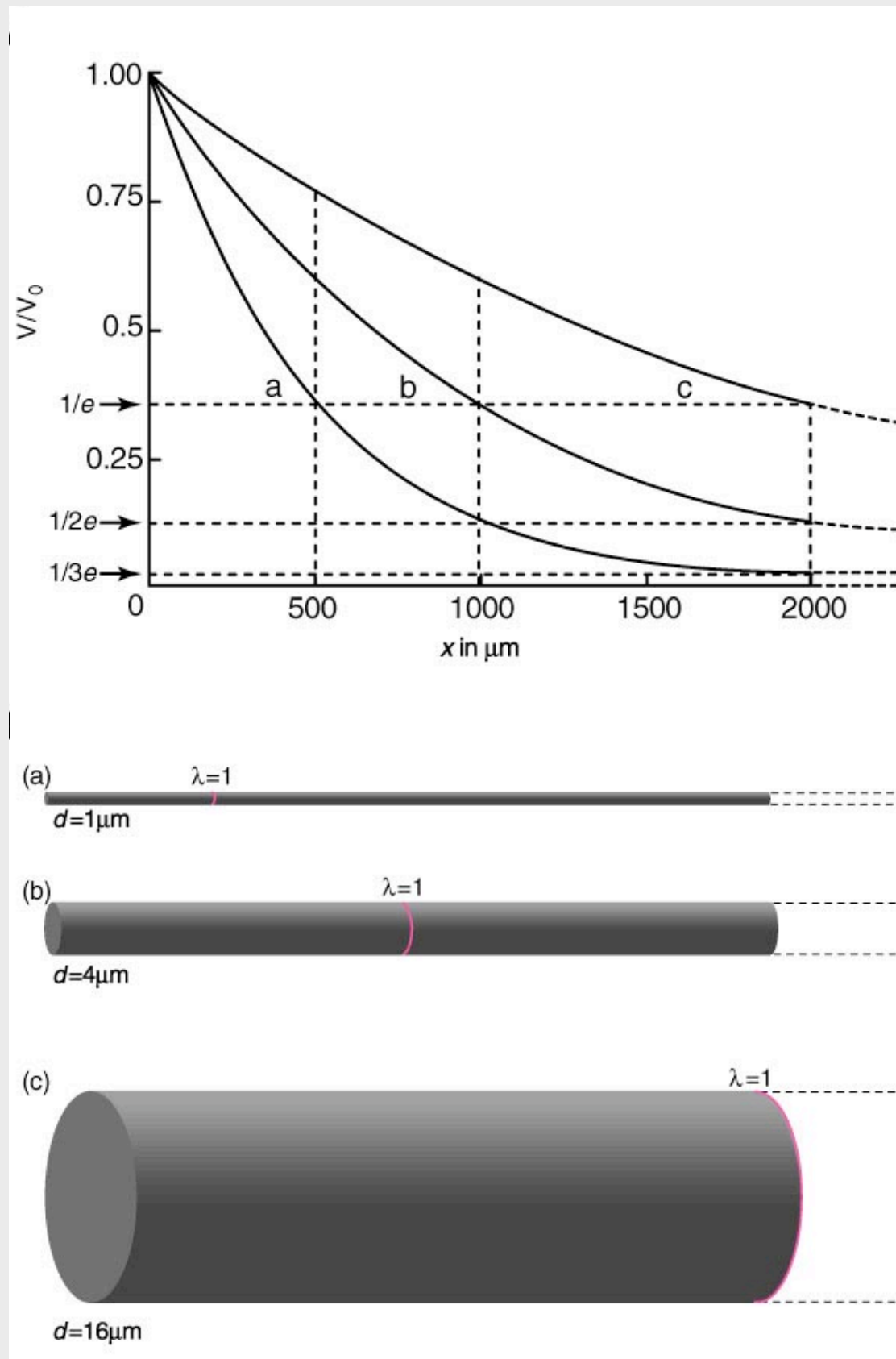
$$V(x) = V_0 \cdot e^{\left(\frac{-x}{\lambda}\right)}$$

$\lambda$  : **space or length constant**

$$\lambda = \sqrt{\frac{R_m}{R_a}}$$

- if  $R_m$  is large (*few leak channels*) the depolarization spreads « further ».
- if  $R_a$  is small (*high ionic mobility in the cylinder*) the depolarization spreads « further ».

## neurons are not isopotential



Ra and therefore  $\lambda$  depends on the diameter of the cylinder

**end of the BREAK**

## axons diameters are variable ...but always small

- squid giant axon : 1 mm

vertebrate :

- peripheral nervous system : 0.2 - 20  $\mu\text{m}$
  - central nervous system : 0.1 - 10  $\mu\text{m}$
- 
- $\lambda$  : 10  $\mu\text{m}$  - 1 mm
  - $\lambda$  : 5  $\mu\text{m}$  - 500  $\mu\text{m}$

length of sciatic nerve ~1m

# saltatory conduction

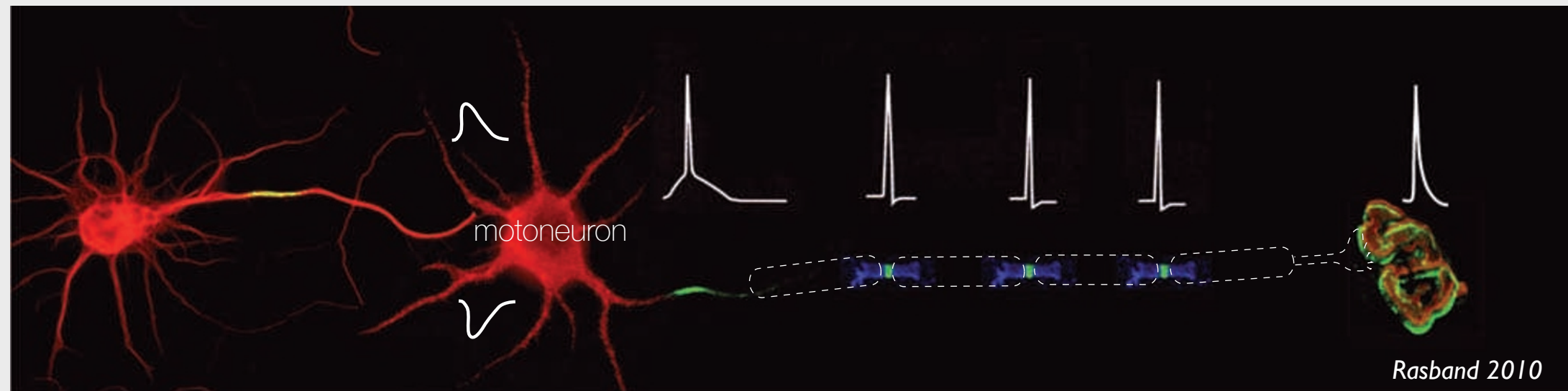
• Subcellular polarity

Synapse

Axon initial segment

Nodes of Ranvier

Neuromuscular junction



- action potential must be rebuilt to propagate

# saltatory conduction

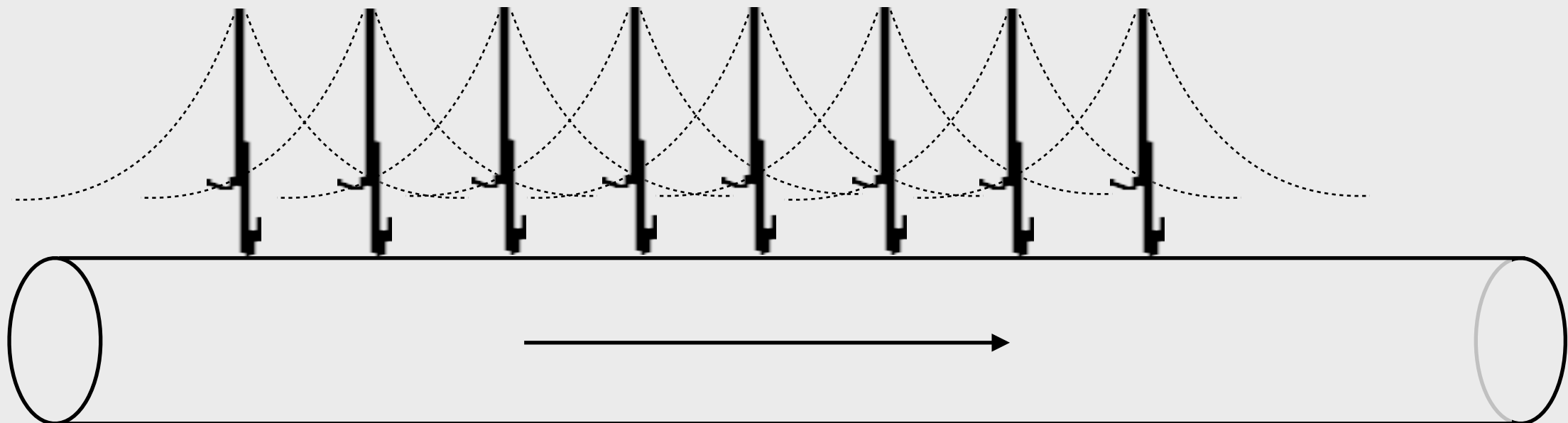
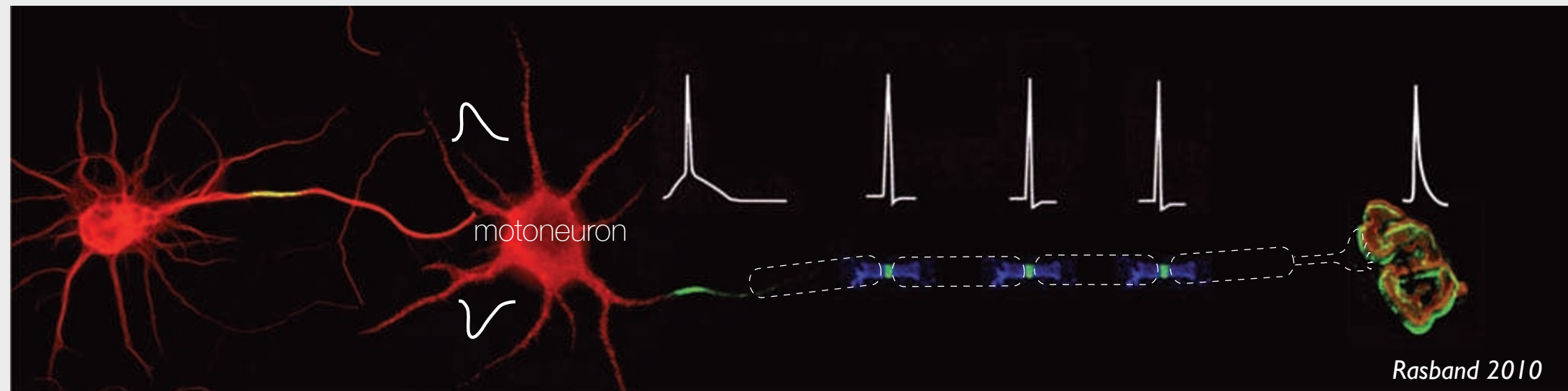
• Subcellular polarity

Synapse

Axon initial segment

Nodes of Ranvier

Neuromuscular junction



- action potential must be rebuilt to propagate
- Na channel inactivation precludes bi-directional propagation

# saltatory conduction

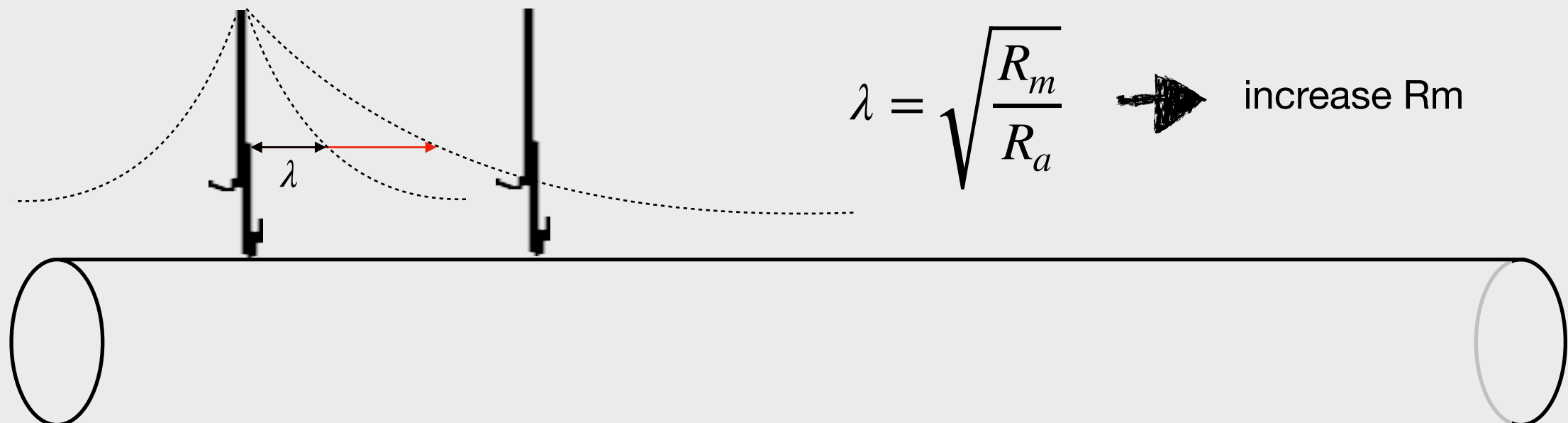
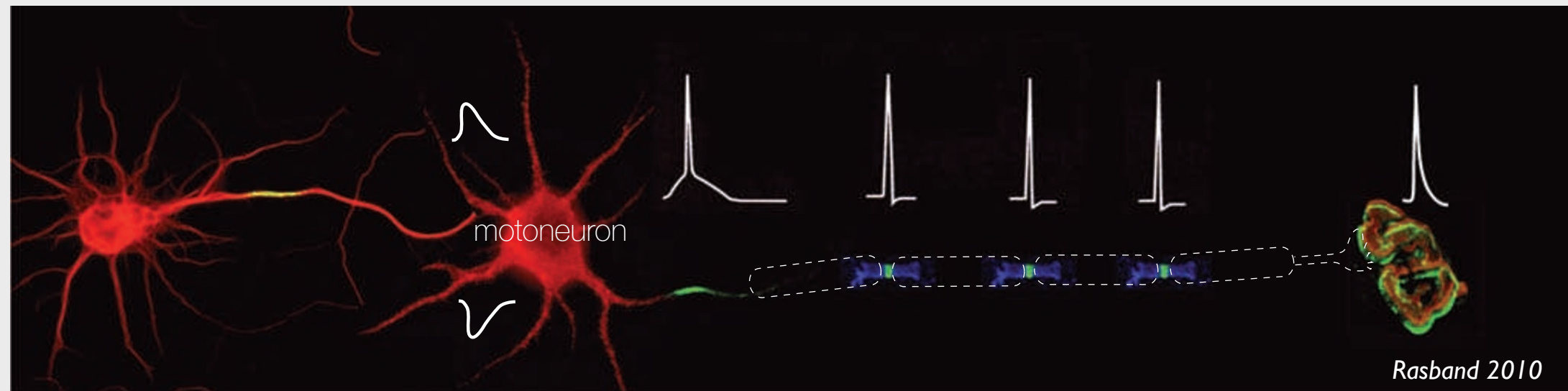
• Subcellular polarity

Synapse

Axon initial segment

Nodes of Ranvier

Neuromuscular junction



# saltatory conduction

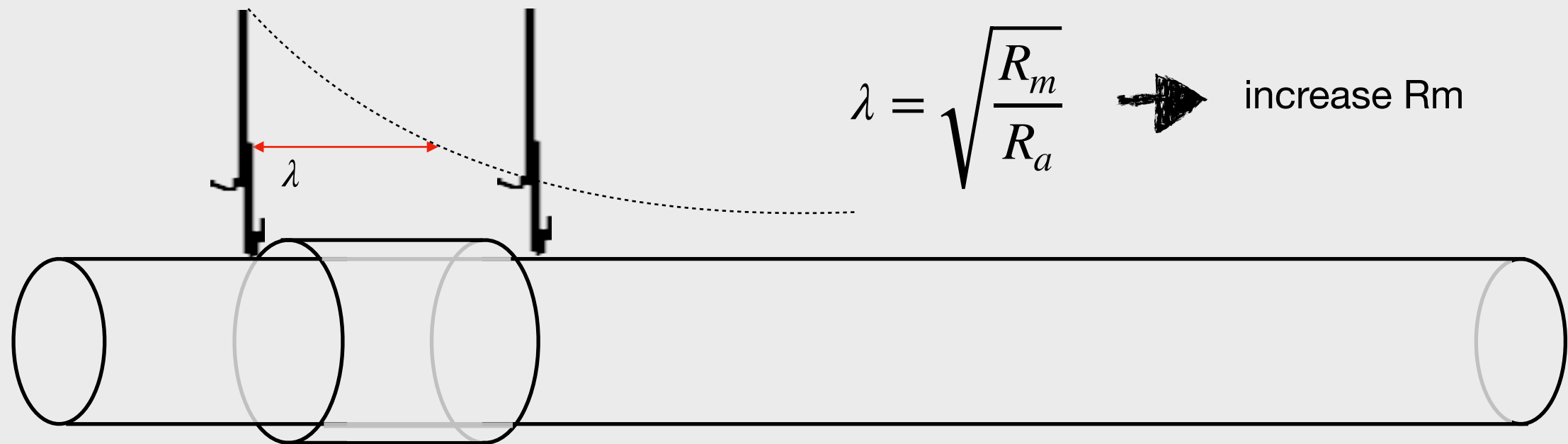
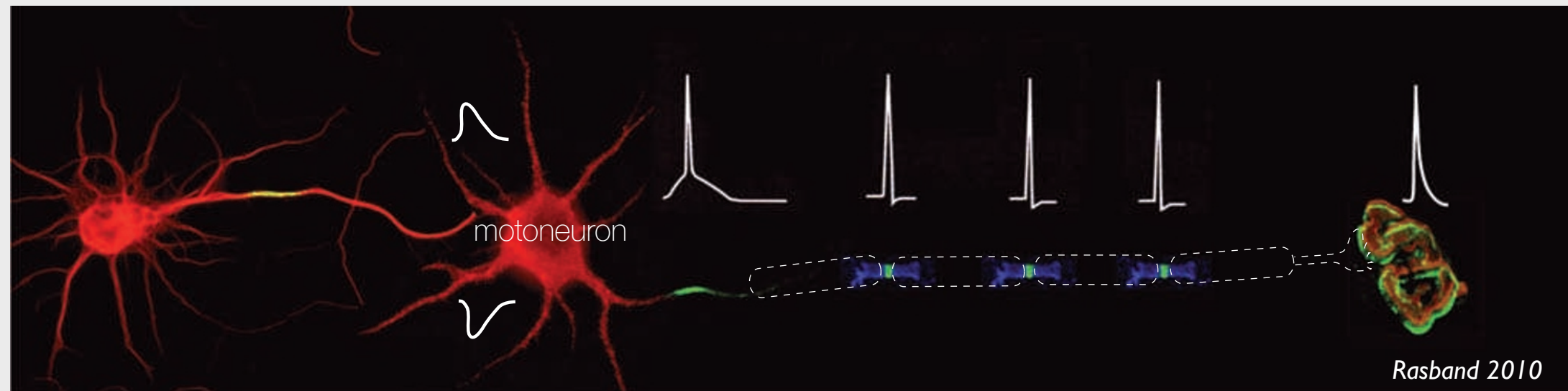
• Subcellular polarity

Synapse

Axon initial segment

Nodes of Ranvier

Neuromuscular junction

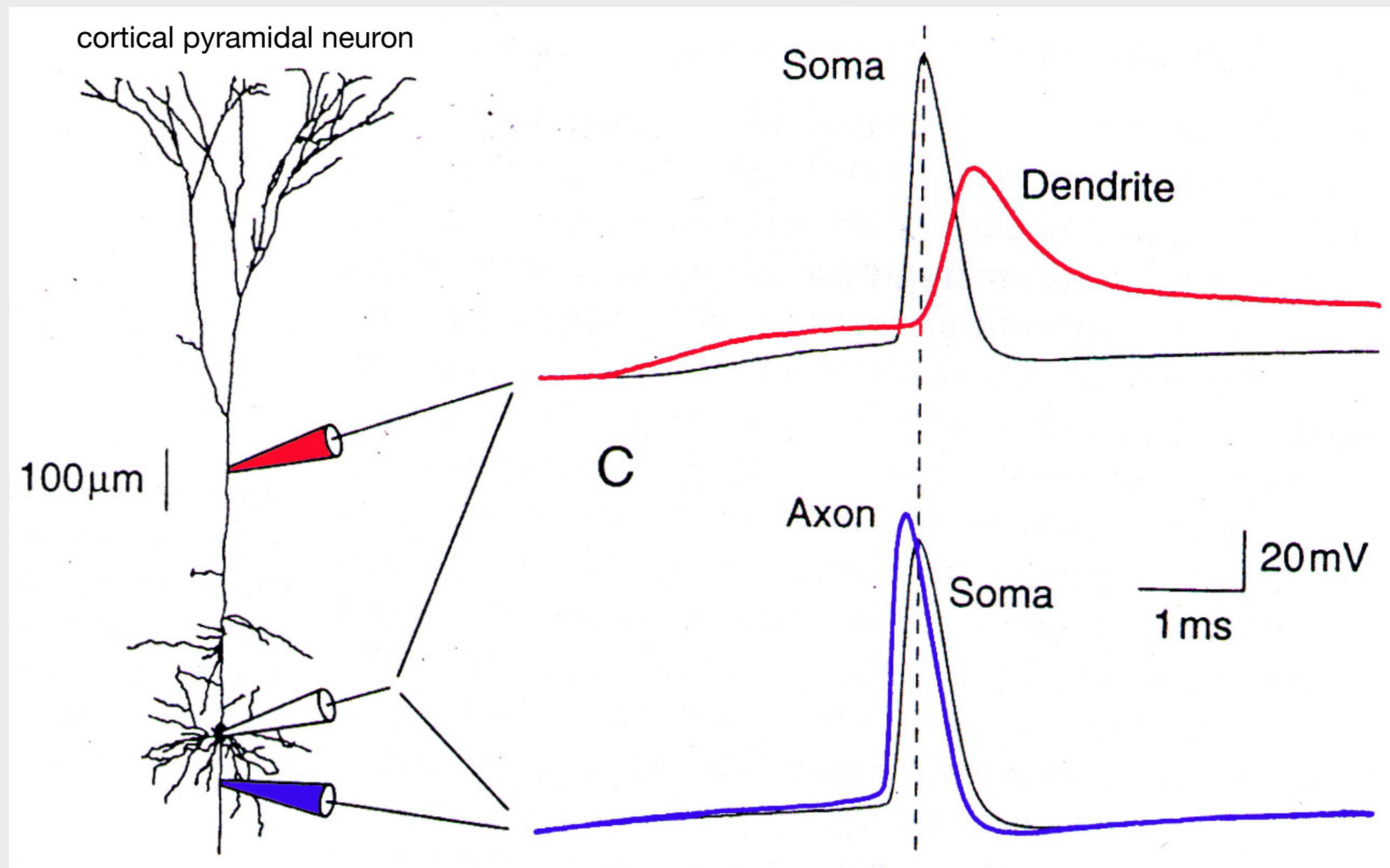


myelin sheaths (Schwann cells in the PNS and oligodendrocytes in the CNS) isolate the axon



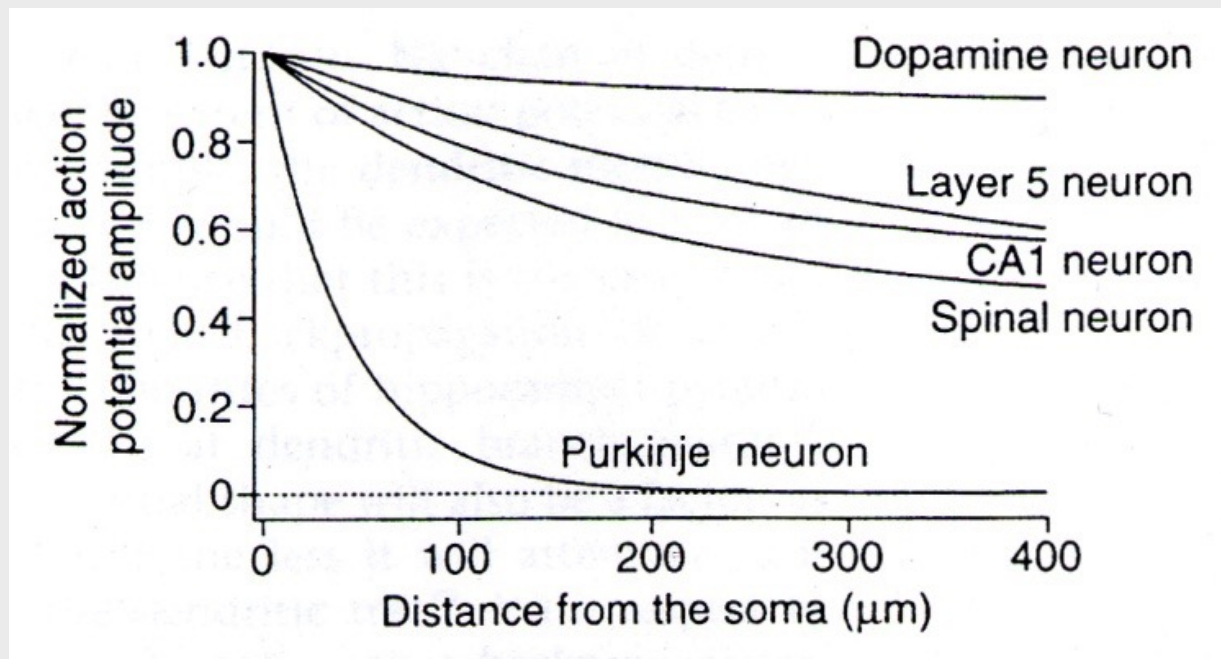
increase  $\lambda$ , increase the conduction speed of the action potential

## action potential backpropagates in the dendrite

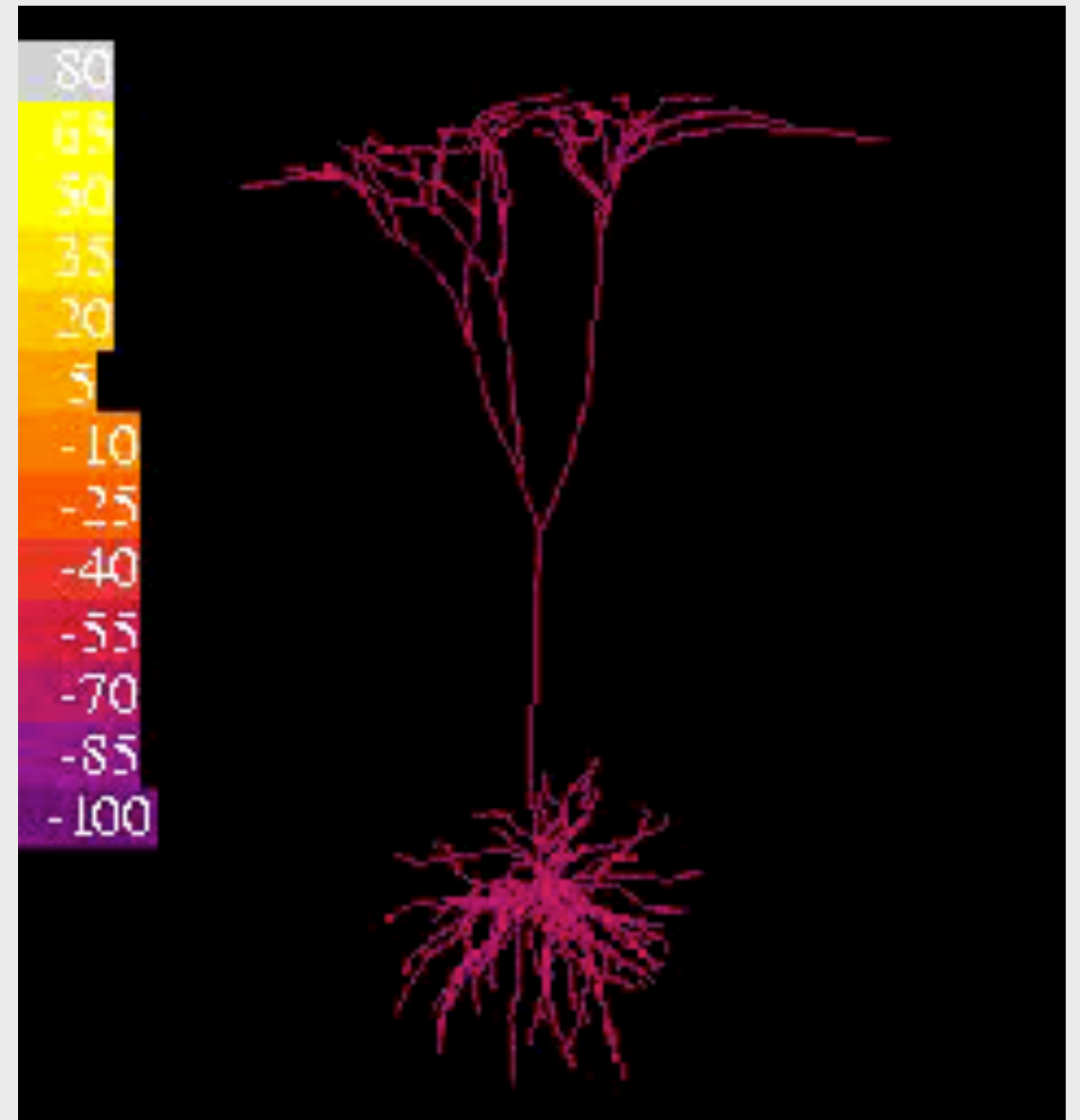


# action potential backpropagates in the dendrite

*distance of backpropagation in different cell populations*

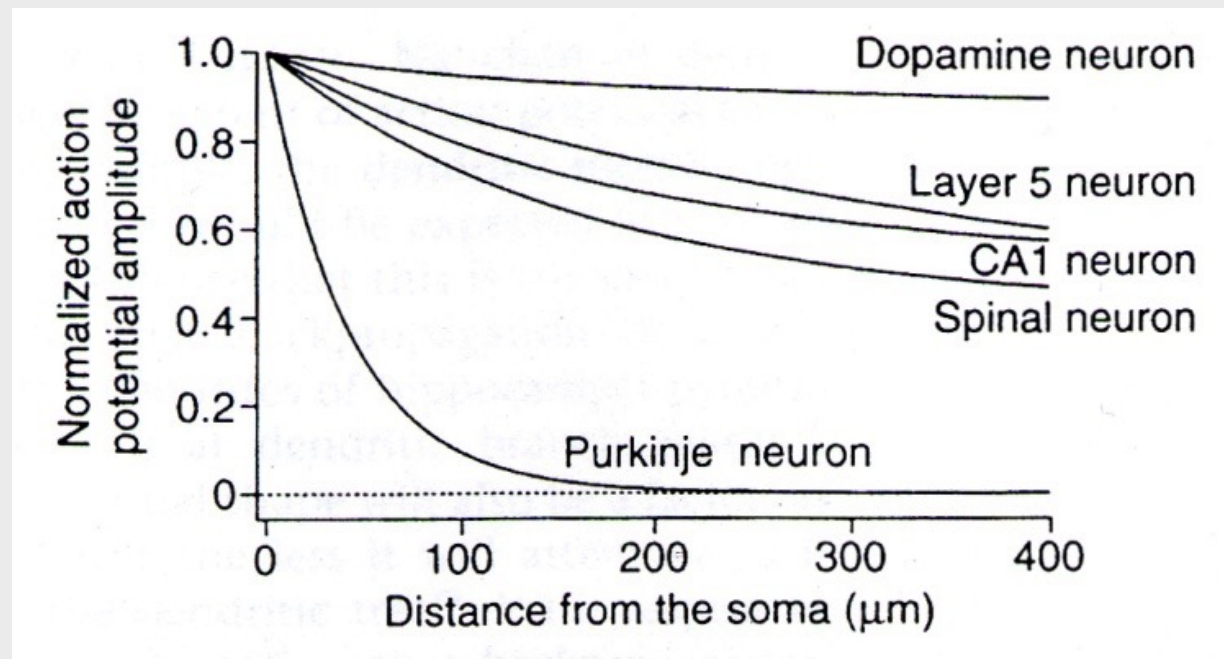
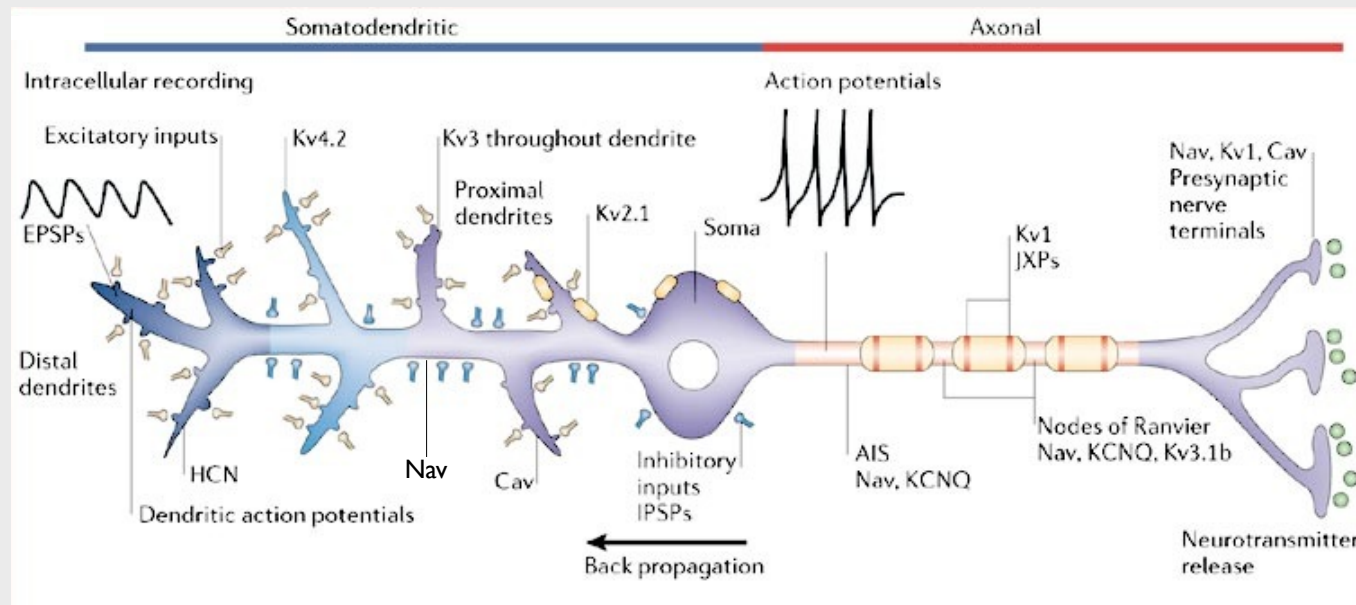


*Stuart et al, 1997*



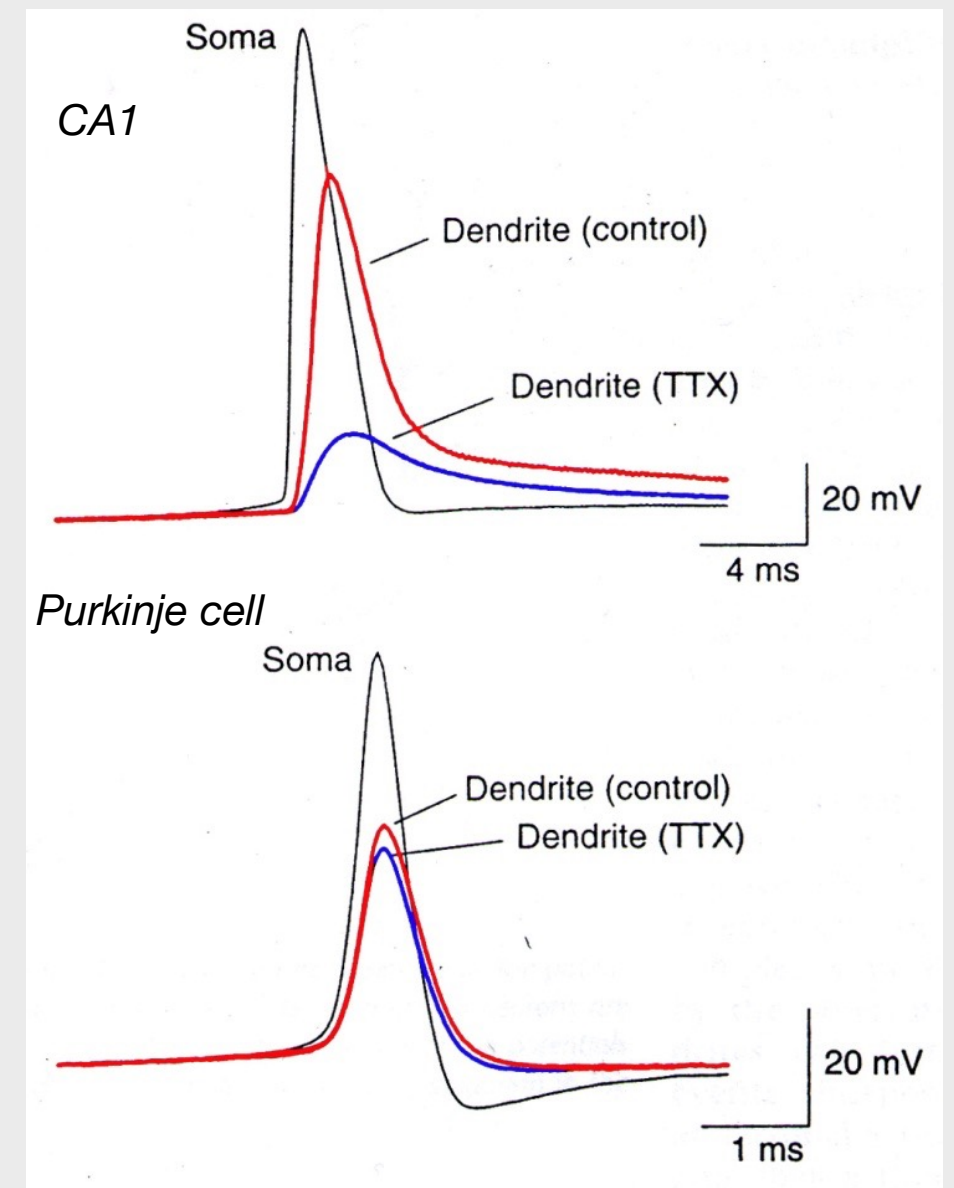
*neocortical layer VI pyramidal neuron (simulation : A. Destexhe)*

# action potential backpropagates in the dendrite

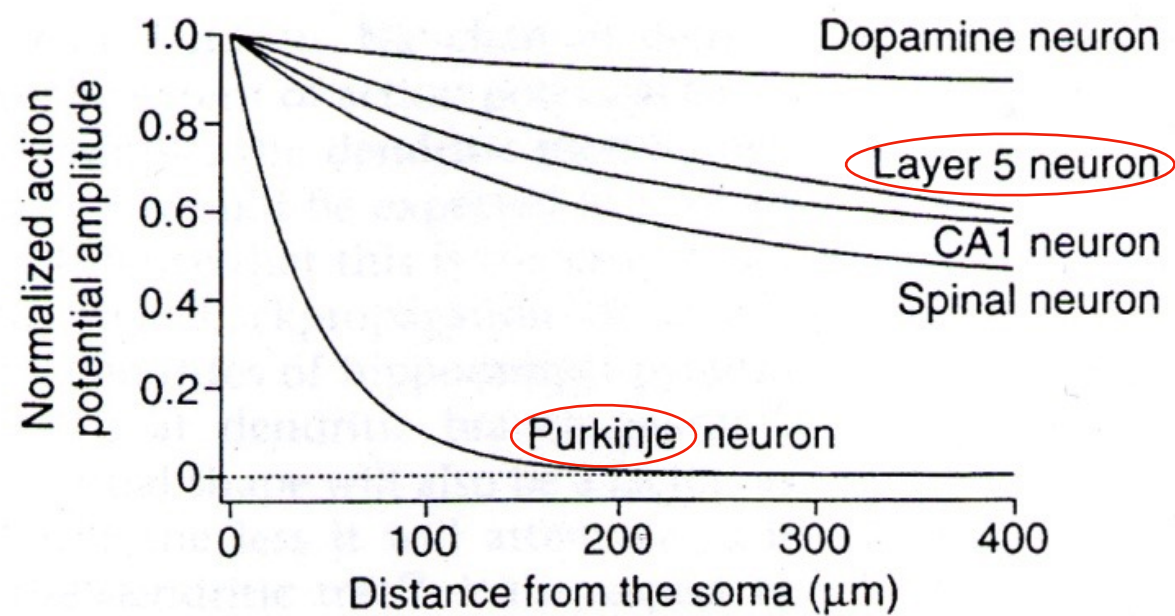
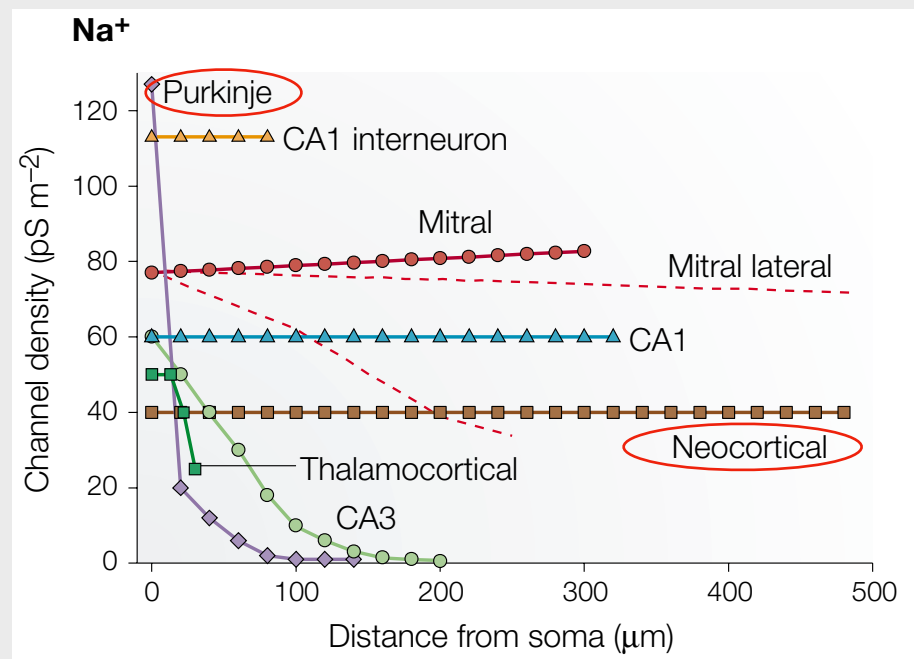


Stuart et al, 1997

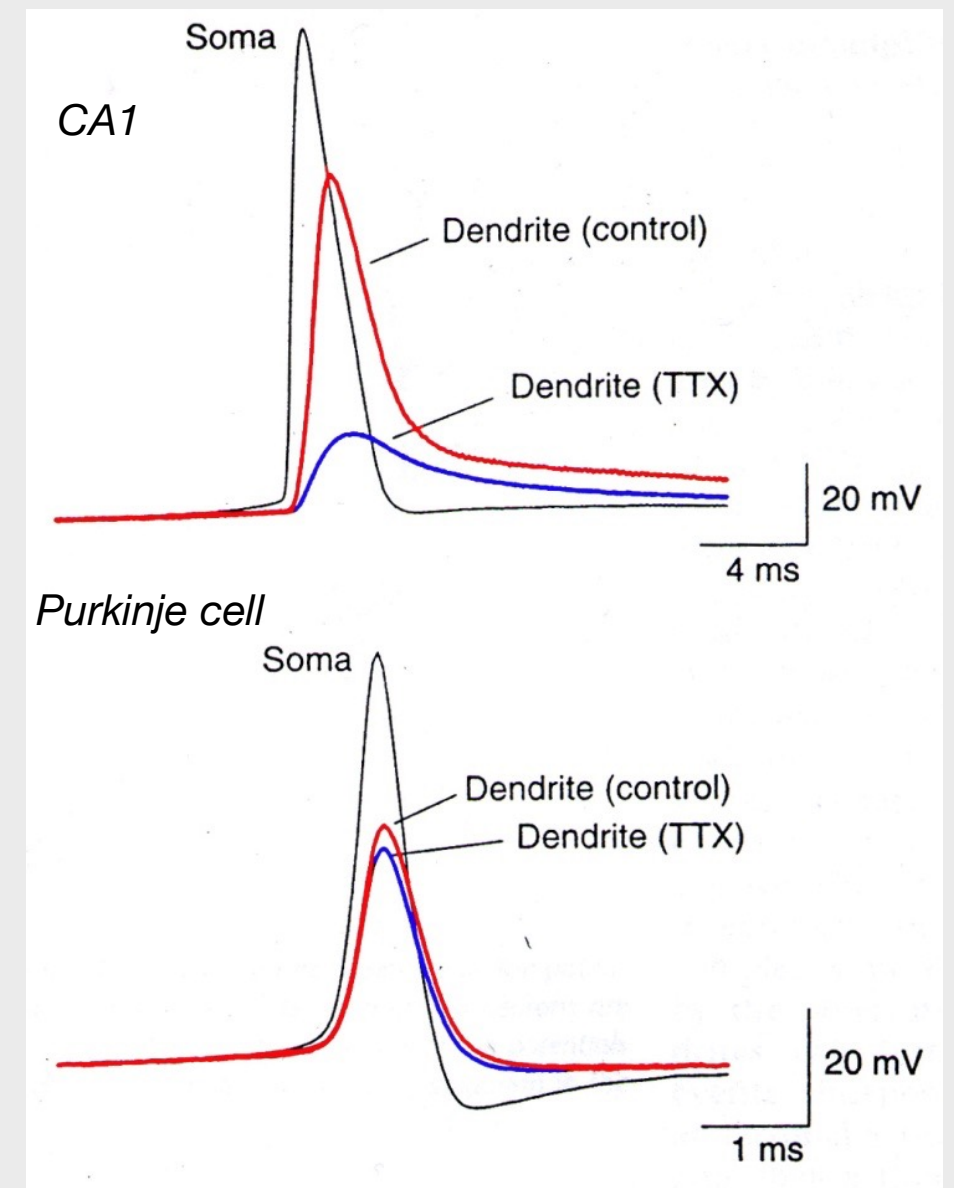
*long distance action potential backpropagation requires Na (and Ca) channels activation in the dendrite*



# action potential backpropagates in the dendrite

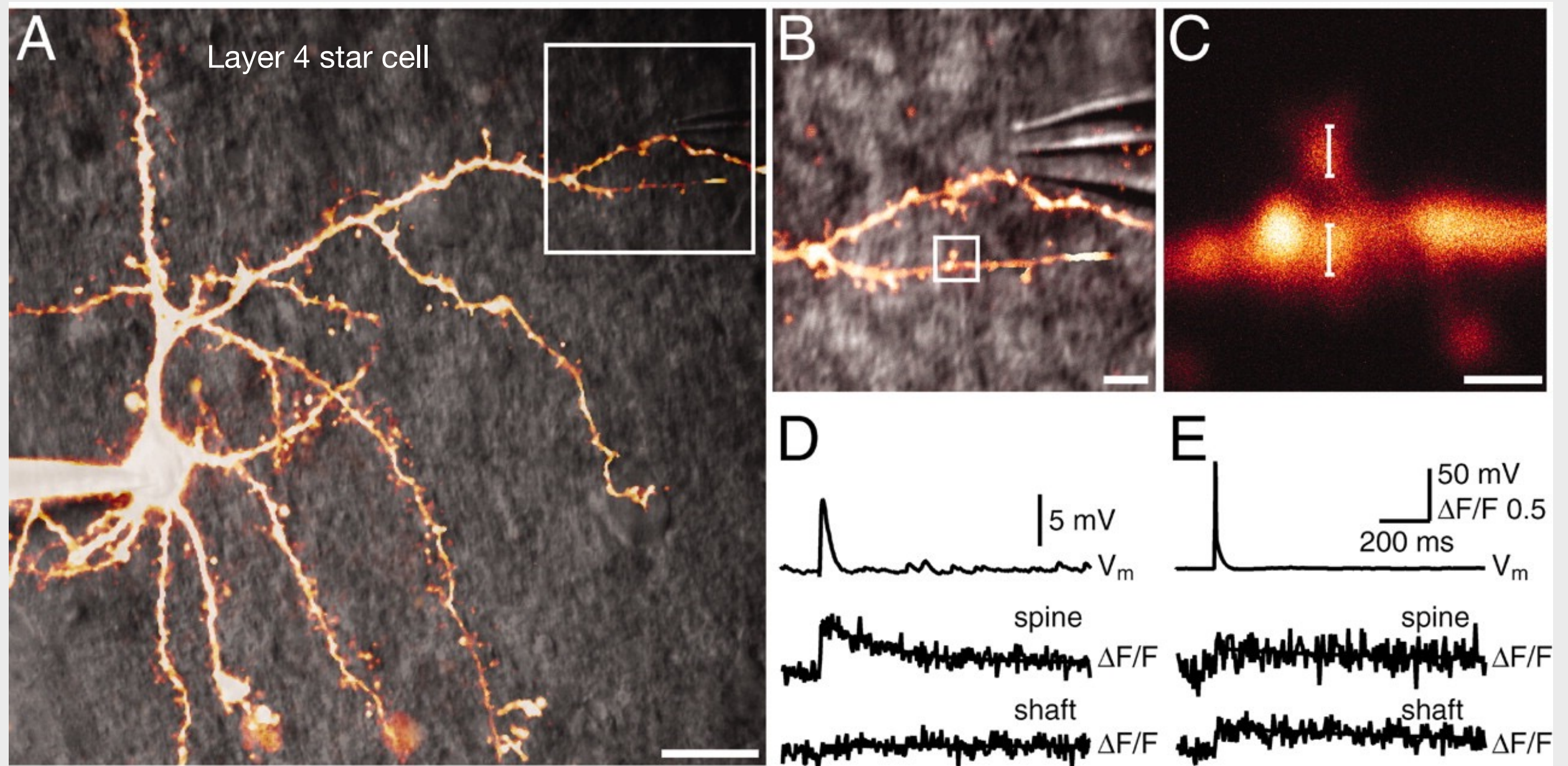


Stuart et al, 1997



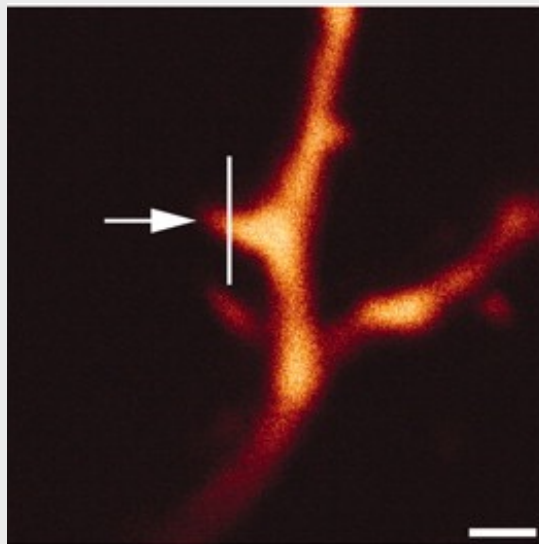
*long distance backpropagation of the action potential requires Na (and Ca) channel activation in the dendrite*

# post-synaptic consequences of action potential backpropagation

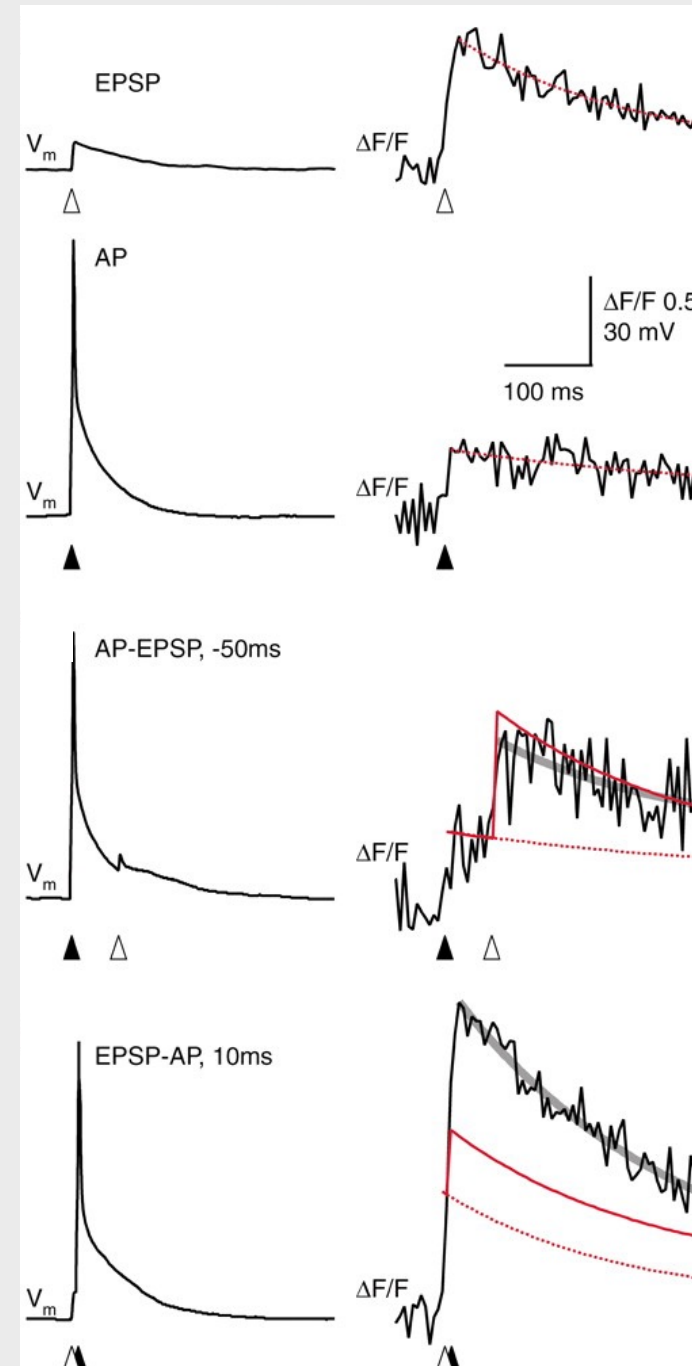


# post-synaptic consequences of action potential backpropagation

Ca<sup>2+</sup> concentration in spine



EPSP alone



action potential (AP) alone

AP followed by an EPSP

EPSP followed by an AP